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CLASS : SIB 2G
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MATKUL : DATA WAREHOUSE (JOBSHEET 3)

Jobsheet 3: Database Analytical

Tujuan Praktikum

Setelah melakukan praktikum ini, mahasiswa diharapkan dapat lebih mengenal table dimensi, table fakta dan apa itu OLAP.

Studi Kasus

Berdasarkan studi kasus sebelumnya, Pimpinan dari LegendVehicle mulai mengenal teknologi dan berpandangan visioner. LegendVehicle akan menerapkan data warehouse pada proses bisnisnya untuk menganalisa proses transaksi yang dilihat dari pembayaran yang masuk. Proses ini digunakan untuk menganalisa KPI "jumlah omset yang dimiliki".

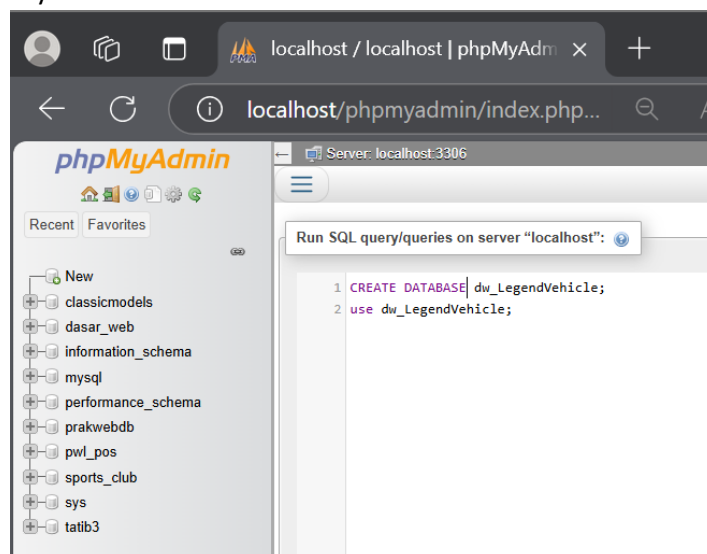
Untuk membentuk data warehouse tersebut, Data Engineer pada LegendVehicle perlu membuat sebuah database yang digunakan sebagai database OLAP.

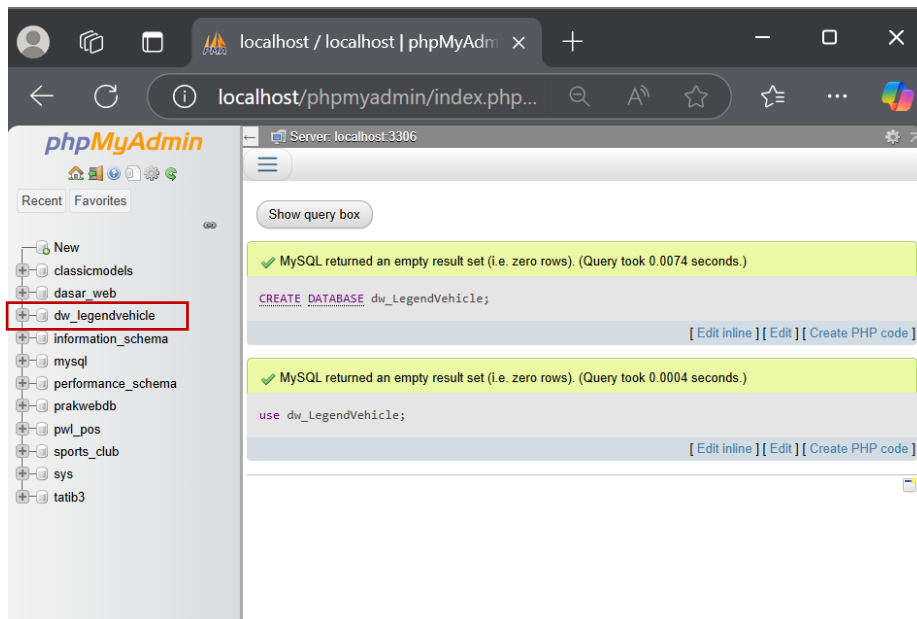
Data dari OLTP yang ada akan di "ETL" kan menuju database OLAP.

A. Dimensi Waktu

1. Buatlah sebuah database yang digunakan sebagai **OLAP** dengan nama **dw_LegendVehicle**.

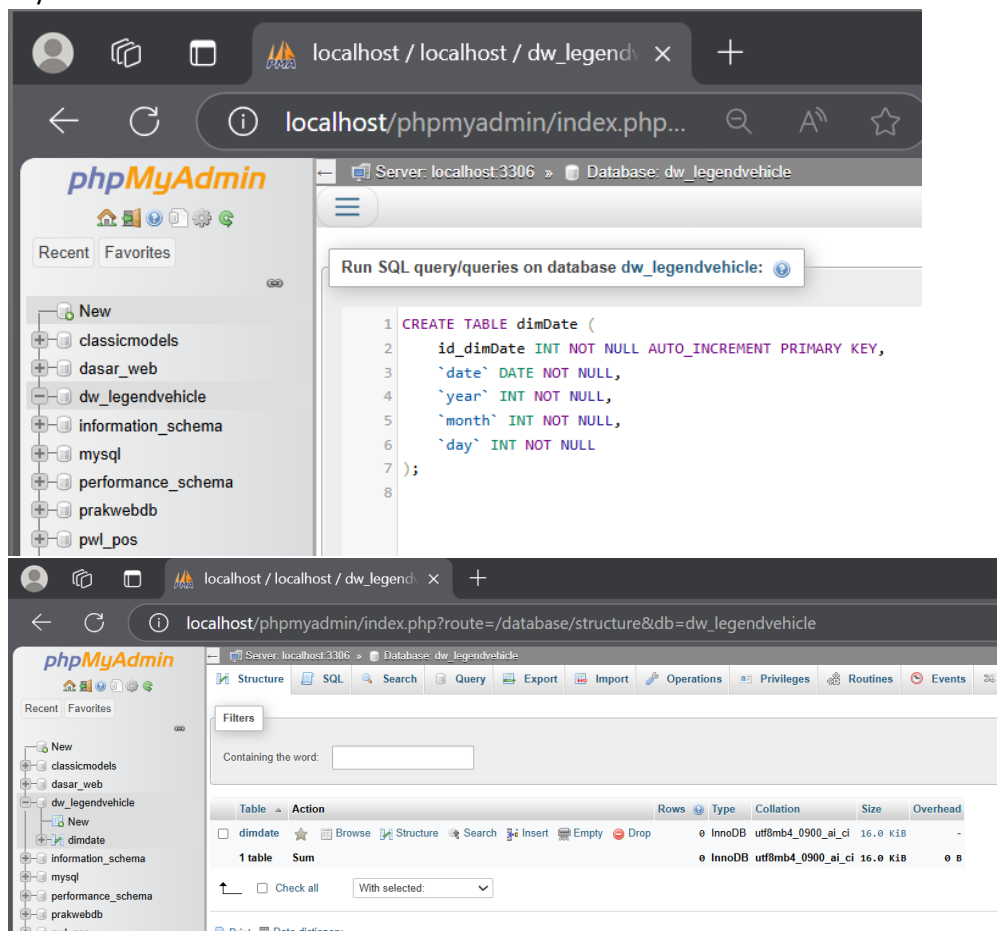
My Answer :





2. Buatlah table untuk menyimpan data master waktu atau yang disebut dengan **tabel dimensi**. Beri nama table tersebut dengan nama **dimDate**.

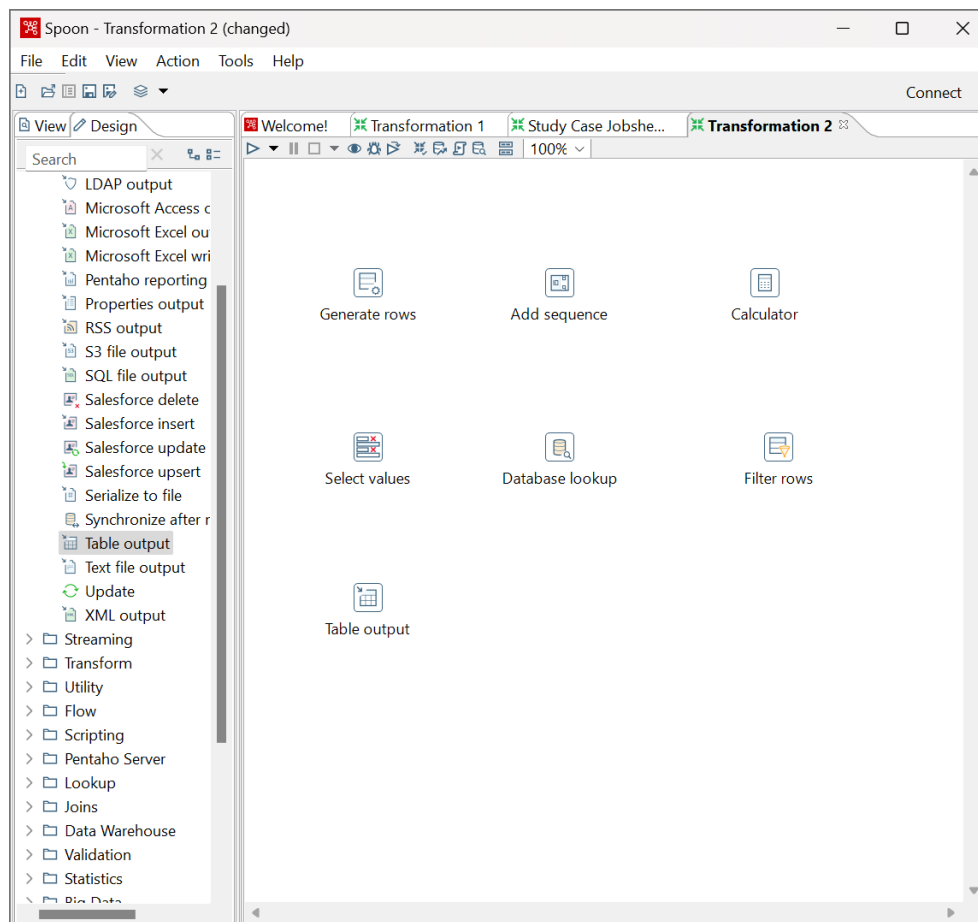
My Answer :



Pada tahapan selanjutnya, untuk membuat tabel dimensi dimDate , maka diperlukan generate data tanggal. Data tanggal yang disiapkan pada tabel dimDate menyesuaikan dengan proses bisnis yang berjalan.

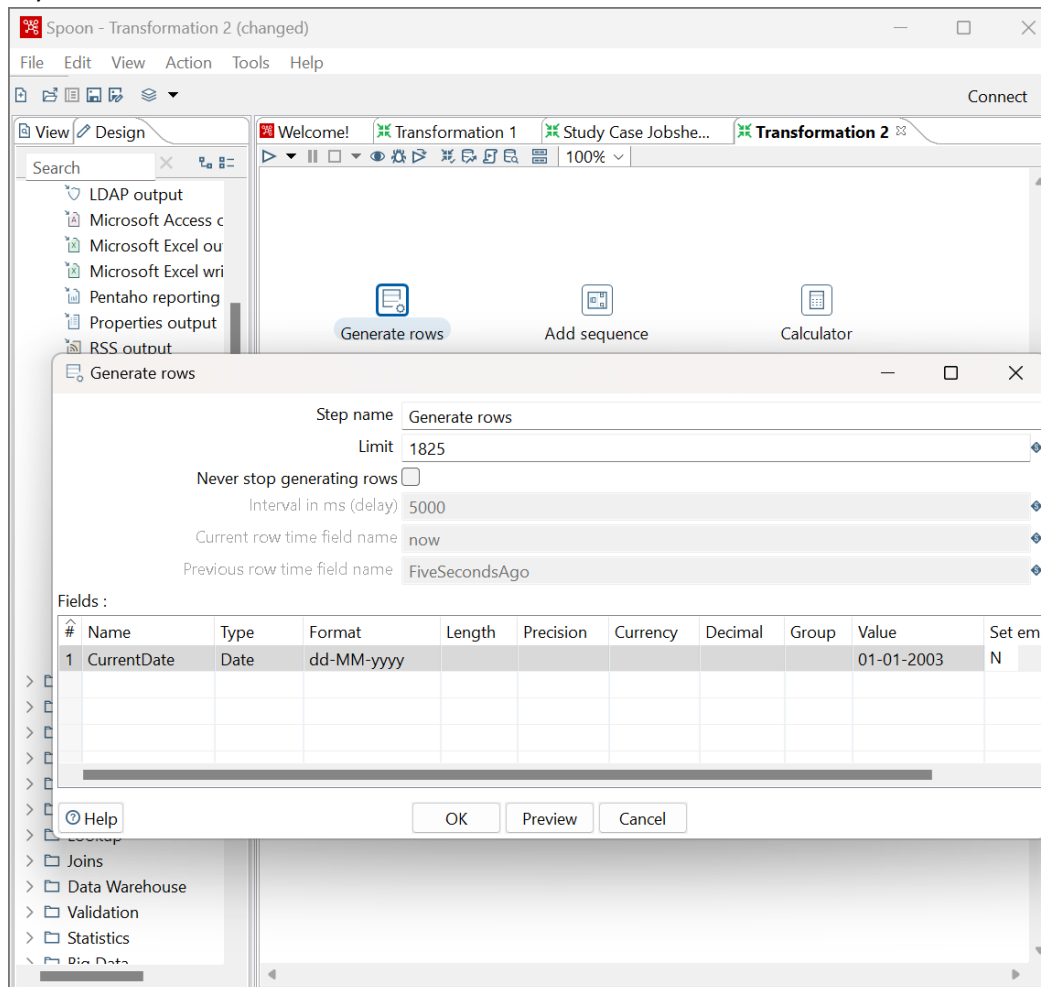
Proses bisnis pada LegendVehicle adalah 5 tahun. Sehingga data pada tabel dimdate yang harus tersedia adalah tanggal selama 5 tahun. Mulai dari 1 Januari 2023

1. Buka PDI Spoon. Buat Transformation baru -> **File - New - Transformation.**
2. Drag and Drop beberapa objek yaitu:
 - **Generate Rows:** digunakan untuk membuat baris data baru.
 - **Add Sequence:** digunakan untuk membuat sequence, dalam hal ini membuat data di setiap harinya.
 - **Calculator:** digunakan untuk menjumlahkan hari dan mengambil data tahun, bulan dan hari.
 - **Select Values:** digunakan untuk memilih field yang digunakan.
 - **Database Lookup:** digunakan untuk melihat dan memastikan bahwa data yang akan dimasukkan kedalam tabel dimDate tidak kembar atau sama dengan data yang ada pada tabel dimDate itu sendiri.
 - **Filter Rows:** digunakan untuk mengambil data yang belum ada pada table dimDate setelah dicek sebelumnya.
 - **Table Output:** digunakan untuk menyimpan data pada tabel tujuan (dimDate).



- Konfigurasi pada **Generate Rows** adalah merubah **limit** menjadi **1825** dimana memiliki arti bahwa data yang akan dibuat sebanyak 1825 data. 1825 merupakan jumlah hari dalam 5 tahun (365 hari x 5 tahun).
- Membuat fields baru bernama **CurrentDate** dengan **type** data **Date** dan **format dd-MM-yyyy** serta **value** awal **01-01-2003**.

My Answer :



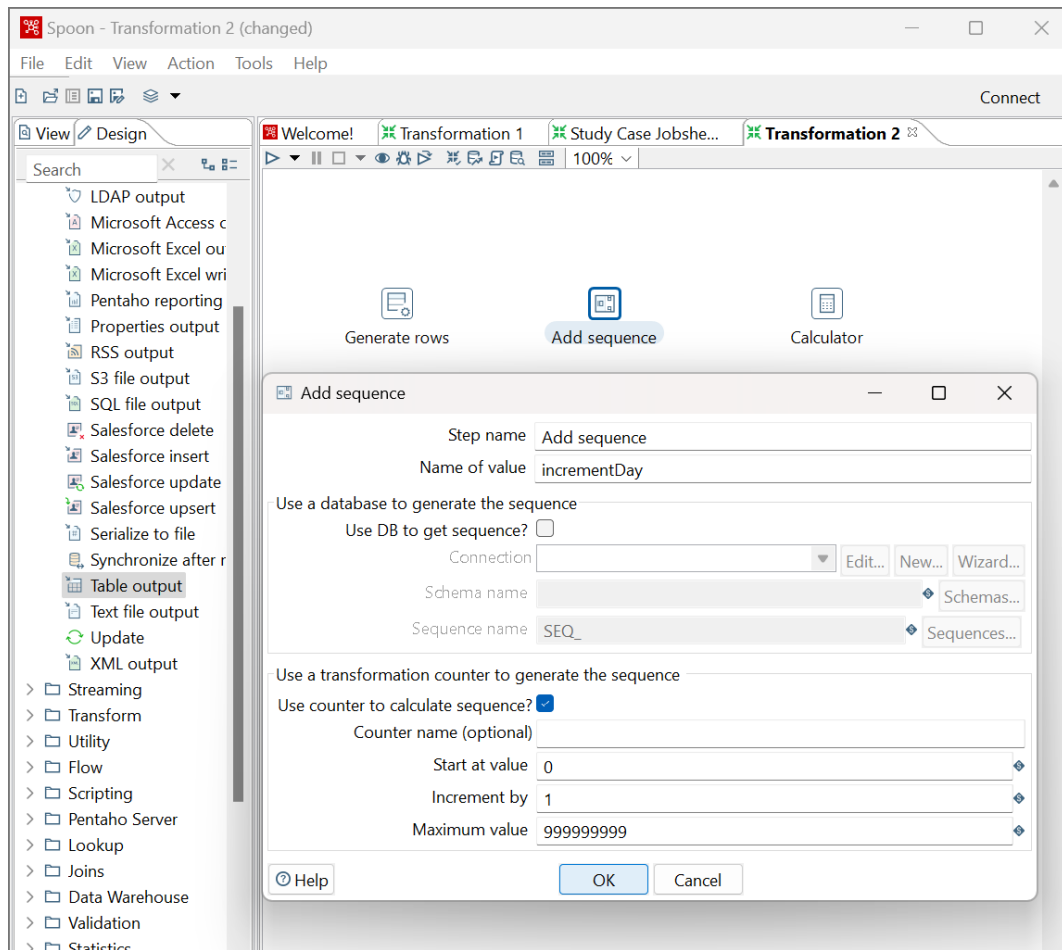
- Hubungkan output dari **Generate Rows** menuju **Add Sequence**.

My Answer :



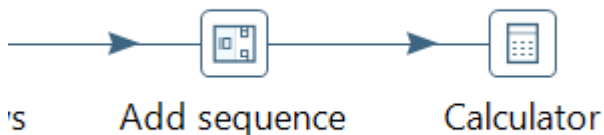
- Konfigurasi pada **Add Sequences** adalah merubah **Name of value** menjadi **incrementDay** dengan **start value** bernilai **0** dan **increment by** bernilai **1**

My Answer :



7. Hubungkan output dari **add sequences** menuju **calculator**.

My Answer :



8. Konfigurasi pada calculator dengan membuat fields baru sebagai berikut:

- **streamDate** merupakan kalkulasi dari **CurrentDate + incrementDay**

Fields:			
#	New field	Calculation	Field A
1	streamDate		
2	streamYear		

CEIL(A)
 FLOOR(A)
 NVL(A, B)
Date A + B Days
 Year of date A
 Month of date A

Fields:					
#	New field	Calculation	Field A	Field B	Field C
1	streamDate	Date A + B Days	CurrentDate	IncrementDay	
2	streamYear				

- **streamYear** merupakan **Year** dari **streamDate**

1	streamDate	Date A + B Days	CurrentDate	Date A + B Days
2	streamYear			Year of date A
3	streamMonth			Month of date A

#	New field	Calculation	Field A	Field B
1	streamDate	Date A + B Days	CurrentDate	IncrementDay
2	streamYear	Year of date A	streamDate	
3	streamMonth	Month of date A	streamDate	
4	streamDay	Day of month of date A	streamDate	

- **streamMonth** merupakan **Month** dari **streamDate**

1	streamDate	Date A + B Days	Date A + B Days
2	streamYear	Year of date A	Year of date A
3	streamMonth		Month of date A
4	streamDay		Day of year of date A

2	streamYear	Year of date A	streamDate
3	streamMonth	Month of date A	streamDate
4	streamDay		

- **streamDay** merupakan **Day of month** dari **streamDate**

Fields:			STDROUND(A , B)
			CEIL(A)
			FLOOR(A)
			NVL(A, B)
			Date A + B Days
			Year of date A
			Month of date A
			Day of year of date A

3	streamMonth	Month of date A	streamDate
4	streamDay	Day of month of date A	streamDate
5			

Calculator

Step name

Calculator

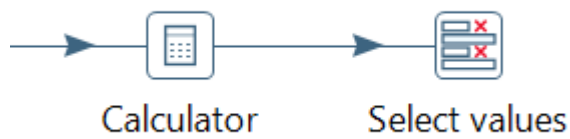
☒ Throw an error on non existing files

Fields:

#	New field	Calculation	Field A	Field B	Field C	Value type	Length	Precision	Remove	Conversion r
1	streamDate	Date A + B Days	CurrentDate	IncrementDay		None			N	
2	streamYear	Year of date A	streamDate			None			N	
3	streamMonth	Month of date A	streamDate			None			N	
4	streamDay	Day of year of date A	streamDate			None			N	

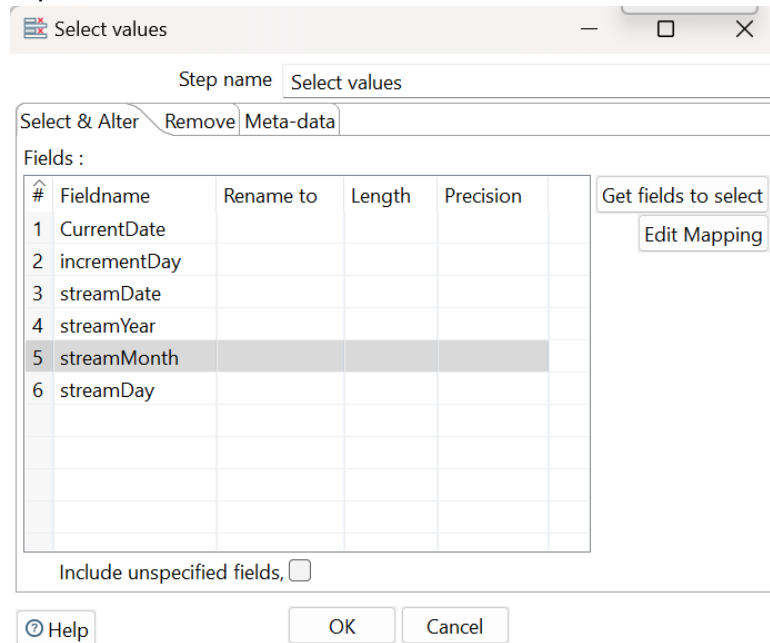
9. Hubungkan output dari **calculator** menuju **Select values**

My Answer :



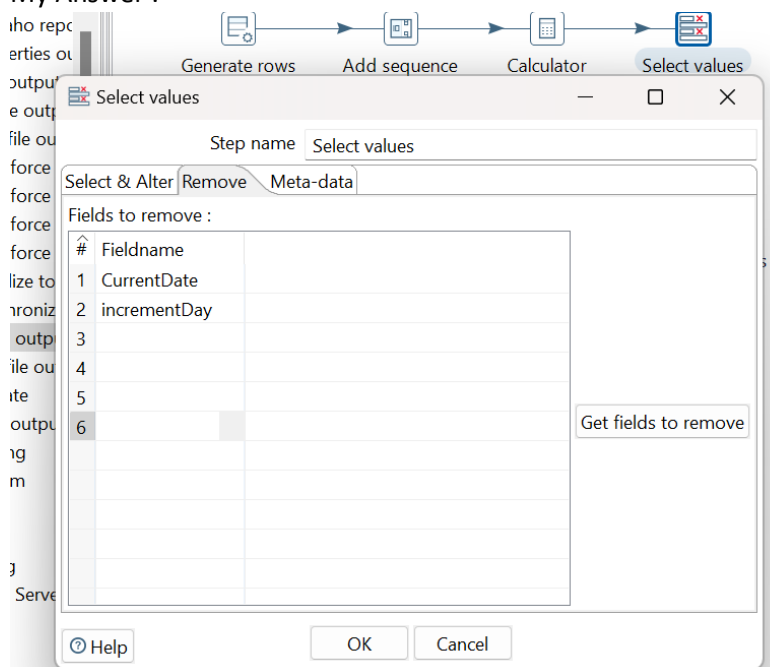
10. Konfigurasi pada **select values** adalah dengan menekan tombol **Get fields to select** pada tab **Select & Alter**. Secara otomatis semua fields dari data input akan muncul.

My Answer :



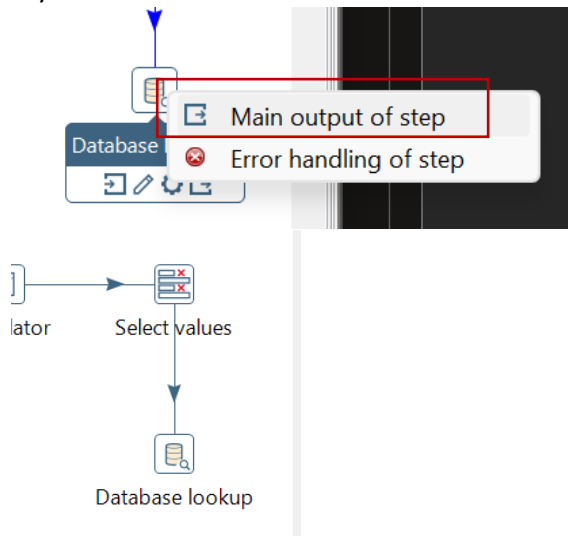
11. Dikarenakan tidak semua fields digunakan, maka pada tab **Remove** diisikan fields **CurrentDate** dan **incrementDay** dikarenakan kedua fields tersebut tidak digunakan.

My Answer :



12. Hubungkan output select values menuju database lookup

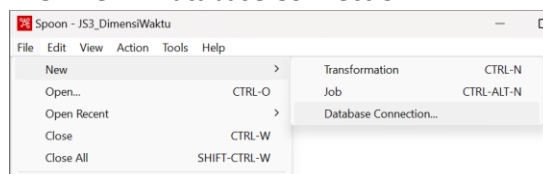
My Answer :



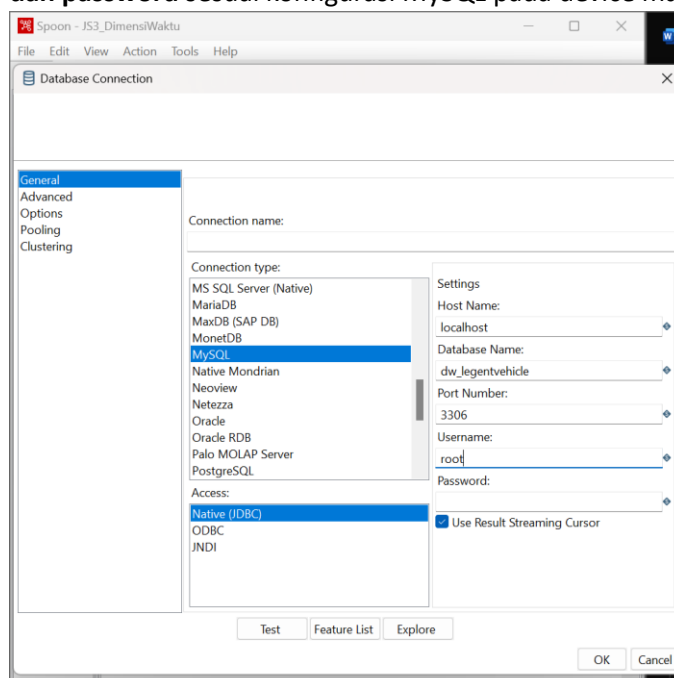
13. Sebelum melakukan konfigurasi pada **database lookup**, buatlah koneksi terlebih dahulu pada database melalui **File - New - Database Connection**. Gunakan **Connection type MySQL** dengan **host name** , **database name**, **port number**, **username** dan **password** sesuai konfigurasi MySQL pada device masing-masing. beri nama **connection name** tersebut dengan nama **conn_dw_destination**.

My Answer :

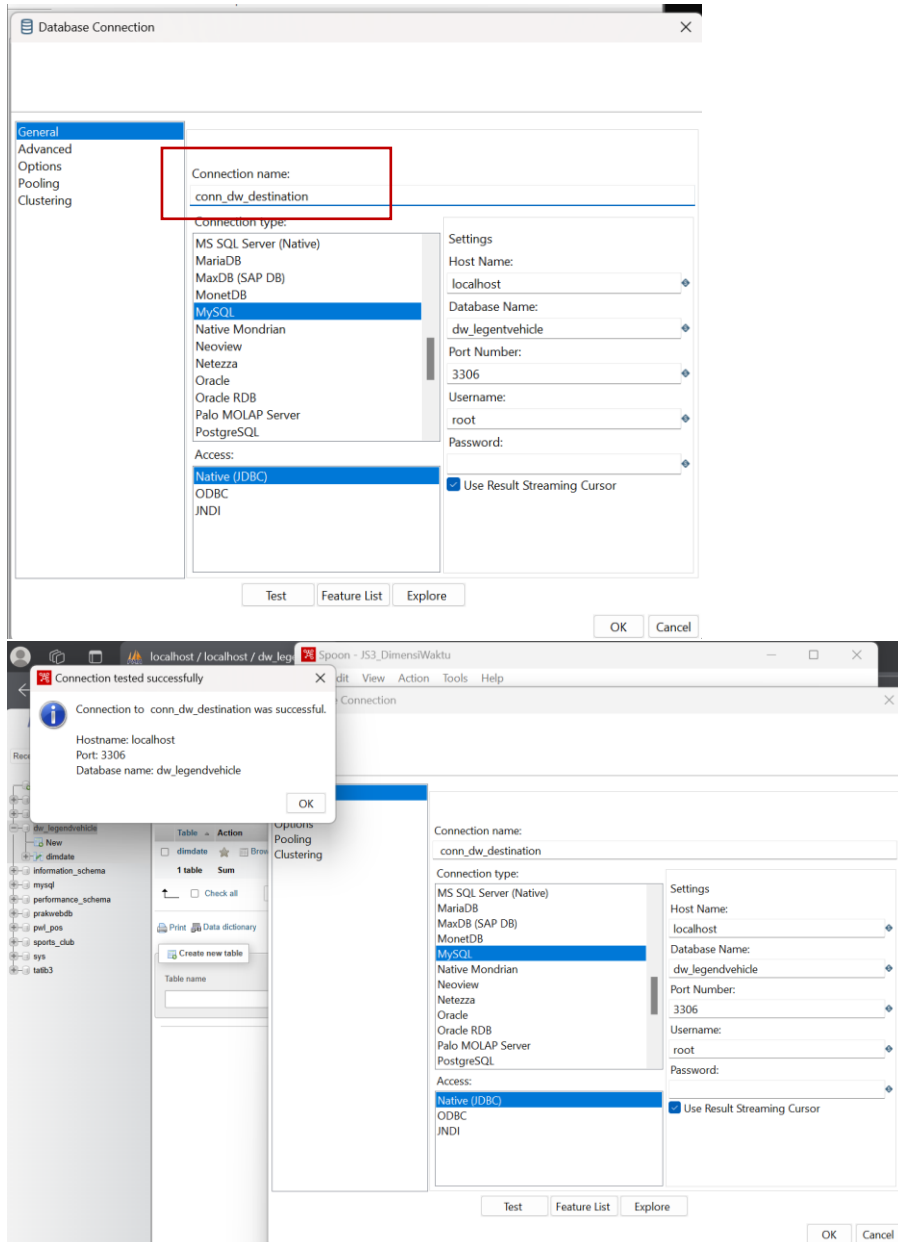
a. **File - New - Database Connection**



b. **Connection type MySQL** dengan **host name**, **database name**, **port number**, **username** dan **password** sesuai konfigurasi MySQL pada device masing-masing.

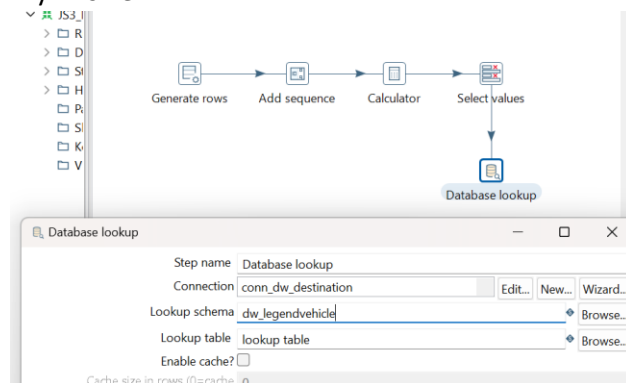


- c. Memberikan nama Connection Name dengan nama “conn_dw_destination”



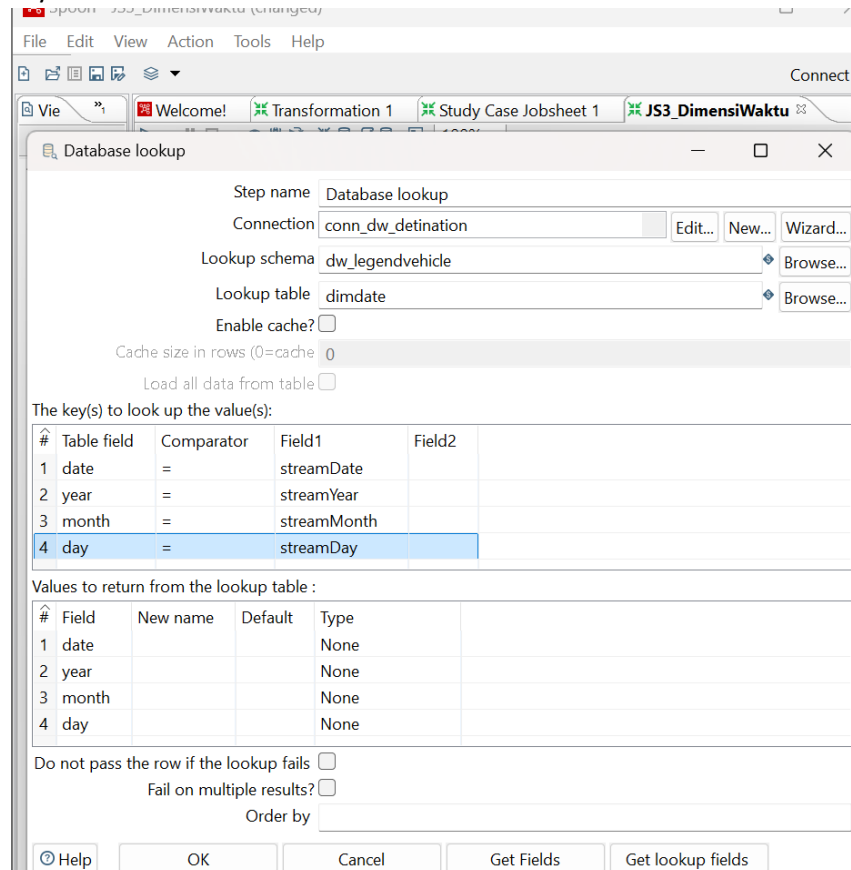
14. Konfigurasi pada **database lookup** adalah dengan memberikan **connection** dengan koneksi yang sudah dibuat pada step sebelumnya. dengan **schema** nama database yang digunakan dan **tabel dimdate** yang telah dibuat pada langkah pertama.

My Answer :



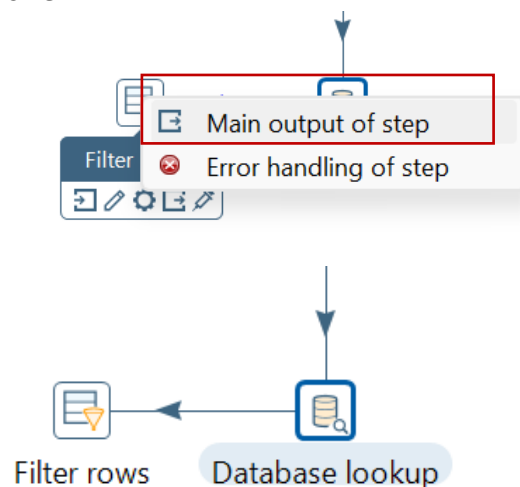
15. Field yang akan dicek untuk melihat kesamaan isi datanya agar tidak kembar adalah:
- field **date** pada table **dimdate** dengan field **streamDate**
 - field **year** pada table **dimdate** dengan field **streamYear**
 - field **month** pada table **dimdate** dengan field **streamMonth**
 - field **day** pada table **dimdate** dengan field **streamDay**
16. Field yang akan di **retrive** adalah field yang ada pada table **dimDate** yaitu **date, year, month, dan day**.

My Answer :



17. Hubungkan output dari **database lookup** dengan **filter rows**

My Answer :



18. Konfigurasi pada **filter rows** adalah dengan melakukan konfigurasi **output true data** pada **table output**. Pada bagian ini data yang tidak memiliki kesamaan pada tahapan sebelumnya akan dicek dimana jika **fields Stream** tidak memiliki kesamaan dengan **field dimDate**, maka **field dimDate** tersebut akan bernilai **null**. Pada pernyataan kondisi tuliskan (**date is null and year is null and month is null and day is null**)

My Answer :

- Date

Select a field:

- streamDate (Date)
- streamYear (Integer)
- streamMonth (Integer)
- streamDay (Integer)
- date (None)**

Select the function:

- =
- <>
- <
- <=
- >
- >=
- REGEXP
- IS NULL**

 Filter rows

Step name: **Filter rows**

Send 'true' data to step:

Send 'false' data to step:

The condition:

Level 1, Select UP to go up one level

date IS NULL

AND

- Year

Select a field:

streamDate (Date)

streamYear (Integer)

streamMonth (Integer)

streamDay (Integer)

date (None)

year (None)

Select the function:

=

<>

<

<=

>

>=

REGEXP

IS NULL

IS NOT NULL

Step name

Filter rows

Send 'true' data to step:

Send 'false' data to step:

The condition:

^^ UP ^^

Level 2, Select UP to go up one level

year IS NULL

AND

null = []

- Month

Select a field:

streamDate	(Date)
streamYear	(Integer)
streamMonth	(Integer)
streamDay	(Integer)
date	(None)
year	(None)
month	(None)

Select the function:

=
<>
<
<=
>
>=
REGEXP
IS NULL
IS NOT NULL

Step name:

Send 'true' data to step:

Send 'false' data to step:

The condition:

☐

Level 3, Select UP to go up one level

month IS NULL

AND

null = []

- Day

Select a field:

streamDate	(Date)
streamYear	(Integer)
streamMonth	(Integer)
streamDay	(Integer)
date	(None)
year	(None)
month	(None)
day	(None)

Select the function:

=
<>
<
<=
>
>=
REGEXP
IS NULL
IS NOT NULL

Filter rows
— □ ×

Step name:

Send 'true' data to step:

Send 'false' data to step:

The condition:

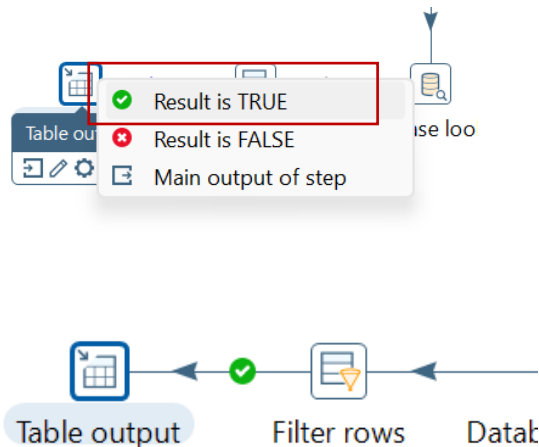
☐
+

```

(
    AND
    (
        AND
        (
            AND
            (
                month IS NULL
                day IS NULL
            )
        )
    )
)
                    
```

19. Hubungkan output dari **filter rows** menuju **table output**.

My Answer :

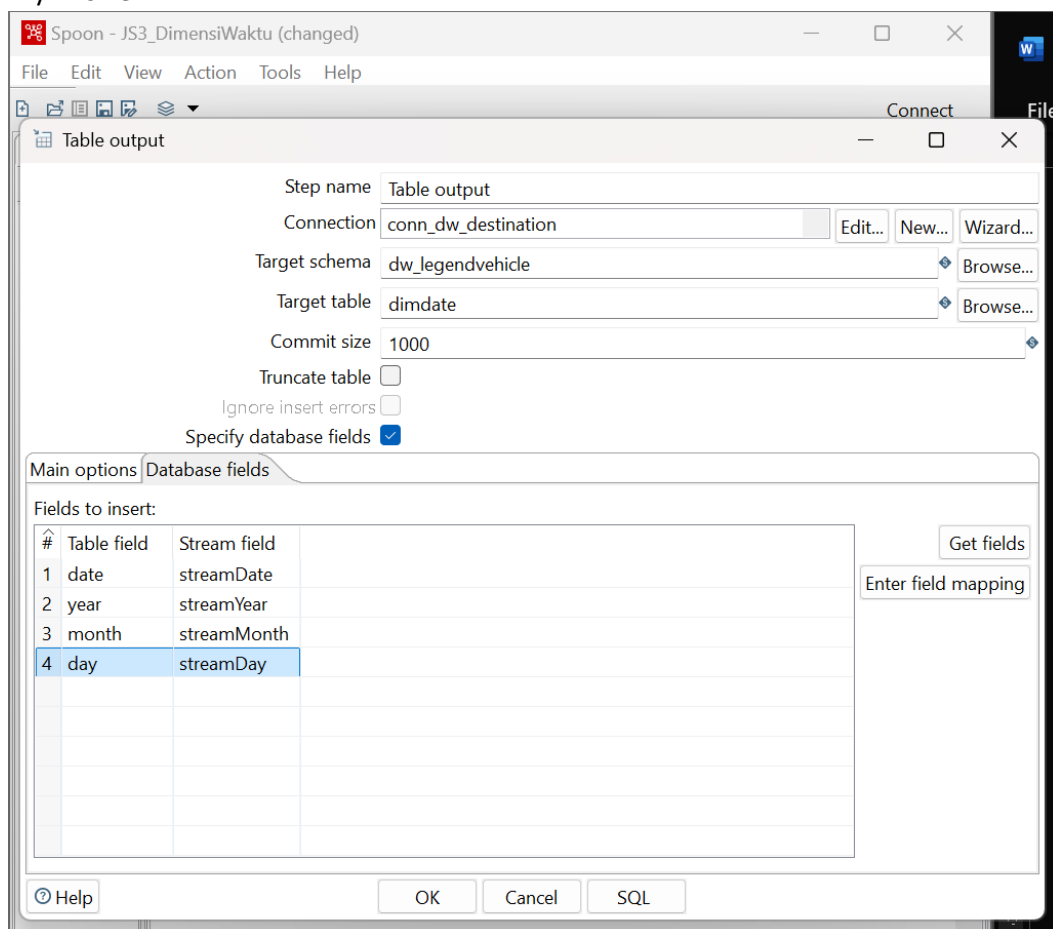


20. Konfigurasi pada **table output** adalah memberikan koneksi pada **conn_dw_destination** dengan **schema dw_legendvehicle** dan table **dimdate**.

21. Aktifkan **specify database fields**.

22. Pada tab **Database fields**, mapping data input **streamDate**, **streamYear**, **streamMonth** dan **streamDay** dengan fields yang ada pada **dimDate**. Pada tahapan ini akan dilakukan insert data menuju tabel **dimDate**.

My Answer :



23. cek isi table dimdate pada database. Jika sukses maka pada table dimdate akan terisi 1825 data.

My Answer :

The image shows two screenshots. The top screenshot is a browser window displaying phpMyAdmin. The URL is localhost/phpmyadmin/index.php?route=/sql&pos=0&db=dw_legendvehic. The interface shows the 'dimdate' table in the 'dw_legendvehic' database. A message at the top states 'Showing rows 0 - 24 (1825 total. Query took 0.0008 seconds.)'. Below this, a table of data is displayed with columns: id_dimDate, date, year, month, and day. The table contains 25 rows of data, starting from 1 2003-01-01 and ending at 25 2003-01-25. The bottom screenshot shows the Spoon ETL job execution log for 'JS3_DimensiWaktu'. The log shows the following steps: Generate rows, Add sequence, Calculator, Select values, Database lookup, Filter rows, and Table output. The log also shows the execution results for each step, including the number of rows processed and the time taken.

localhost / localhost / dw_legendvehic

localhost/phpmyadmin/index.php?route=/sql&pos=0&db=dw_legendvehic

phpMyAdmin

Server: localhost:3306 Database: dw_legendvehic Table: dimdate

Showing rows 0 - 24 (1825 total. Query took 0.0008 seconds.)

SELECT * FROM 'dimdate'

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	id_dimDate	date	year	month	day
☐	1	2003-01-01	2003	1	1
☐	2	2003-01-02	2003	1	2
☐	3	2003-01-03	2003	1	3
☐	4	2003-01-04	2003	1	4
☐	5	2003-01-05	2003	1	5
☐	6	2003-01-06	2003	1	6
☐	7	2003-01-07	2003	1	7
☐	8	2003-01-08	2003	1	8
☐	9	2003-01-09	2003	1	9
☐	10	2003-01-10	2003	1	10
☐	11	2003-01-11	2003	1	11
☐	12	2003-01-12	2003	1	12
☐	13	2003-01-13	2003	1	13
☐	14	2003-01-14	2003	1	14
☐	15	2003-01-15	2003	1	15
☐	16	2003-01-16	2003	1	16
☐	17	2003-01-17	2003	1	17
☐	18	2003-01-18	2003	1	18
☐	19	2003-01-19	2003	1	19
☐	20	2003-01-20	2003	1	20
☐	21	2003-01-21	2003	1	21
☐	22	2003-01-22	2003	1	22
☐	23	2003-01-23	2003	1	23
☐	24	2003-01-24	2003	1	24
☐	25	2003-01-25	2003	1	25

Check all With selected: Edit Copy Delete Export

1 > >> Number of rows: 25 Filter rows: Search this table Sort by key: None

Console / SQL operations

Spoon - JS3_DimensiWaktu (changed)

File Edit View Action Tools Help

View Design

Search

Transformations

- JS3_DimensiWaktu
 - Run configuration
 - Database connect
 - Steps
 - Hops
 - Partition schemas
 - Slave server
 - Kettle cluster sche
 - VFS Connections

Generate rows Add sequence Calculator Select values Database lookup Filter rows Table output

Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

2025/03/21 14:39:05 - Spoon - Transformation opened.

2025/03/21 14:39:05 - Spoon - Launching transformation [JS3_DimensiWaktu]...

2025/03/21 14:39:05 - Spoon - Started the transformation execution.

2025/03/21 14:39:05 - JS3_DimensiWaktu - Dispatching started for transformation [JS3_DimensiWaktu]

2025/03/21 14:39:05 - Table output.0 - Connected to database [conn_dw_destination] (commit=1000)

2025/03/21 14:39:05 - Generate rows.0 - Finished processing (I=0, O=0, R=1825, W=1825, U=0, E=0)

2025/03/21 14:39:05 - Add sequence.0 - Finished processing (I=0, O=0, R=1825, W=1825, U=0, E=0)

2025/03/21 14:39:05 - Calculator.0 - Finished processing (I=0, O=0, R=1825, W=1825, U=0, E=0)

2025/03/21 14:39:05 - Select values.0 - Finished processing (I=0, O=0, R=1825, W=1825, U=0, E=0)

2025/03/21 14:39:06 - Database lookup.0 - Finished processing (I=0, O=0, R=1825, W=1825, U=0, E=0)

2025/03/21 14:39:06 - Filter rows.0 - Finished processing (I=0, O=0, R=1825, W=1825, U=0, E=0)

2025/03/21 14:39:07 - Table output.0 - Finished processing (I=0, O=1825, R=1825, W=1825, U=0, E=0)

2025/03/21 14:39:07 - Spoon - The transformation has finished!!

TUGAS 1

1. Buka preview tab pada execution result area di setiap proses object. amati input dan output data yang ada. bandingkan di setiap prosesnya. jelaskan perbedaan di setiap prosesnya.

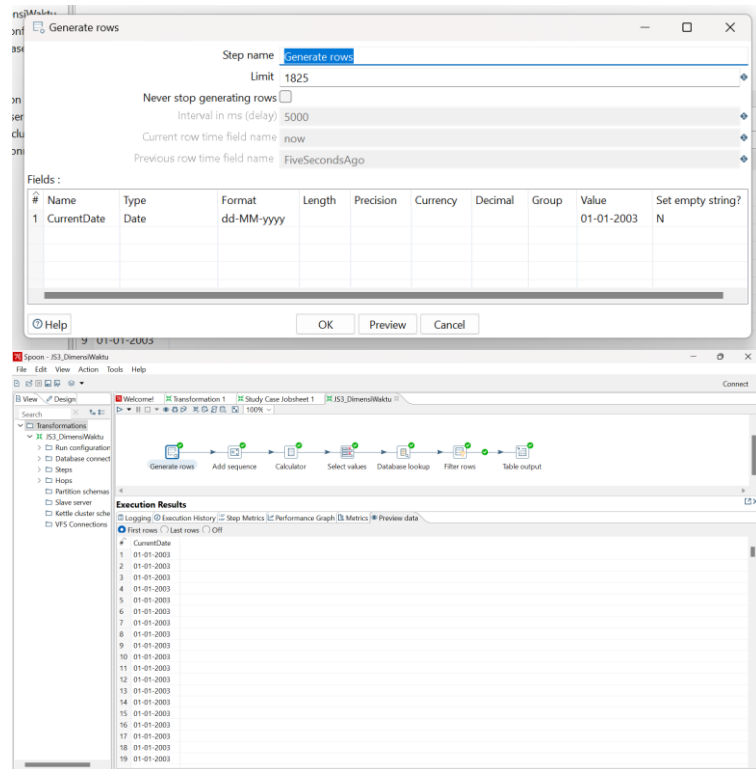
Generate Rows

Keterangan :

Generate Rows pada tabel di samping digunakan untuk membuat data secara otomatis dalam jumlah tertentu sesuai kebutuhan.

CurrentDate → berguna untuk tipe data Date dengan format dd-MM-yyyy. Nilai yang keluar darinya adalah 01-01-2003.

SS Data Input dan Output :

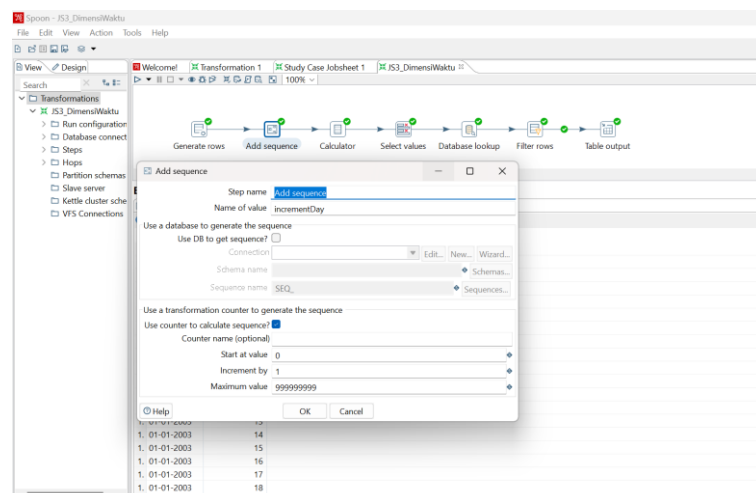


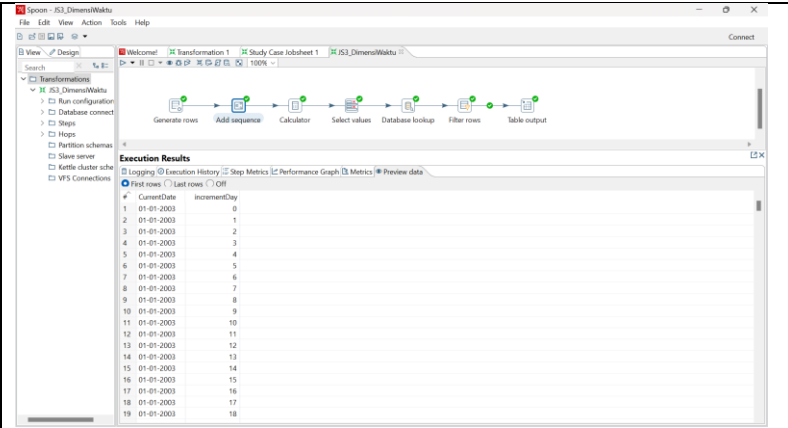
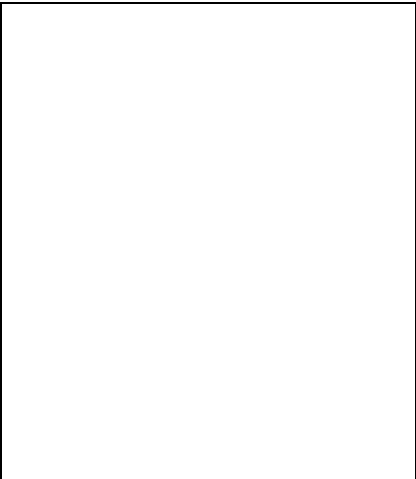
Add Sequences

Keterangan :

Add Sequences berguna untuk menambahkan nomor Sequences pada data yang telah dibuat. Pada gambar, Add Sequences menambahkan kolom baru, IncrementDay, dengan nilai angka yang meningkat secara bertahap.

SS Data Input dan Output :

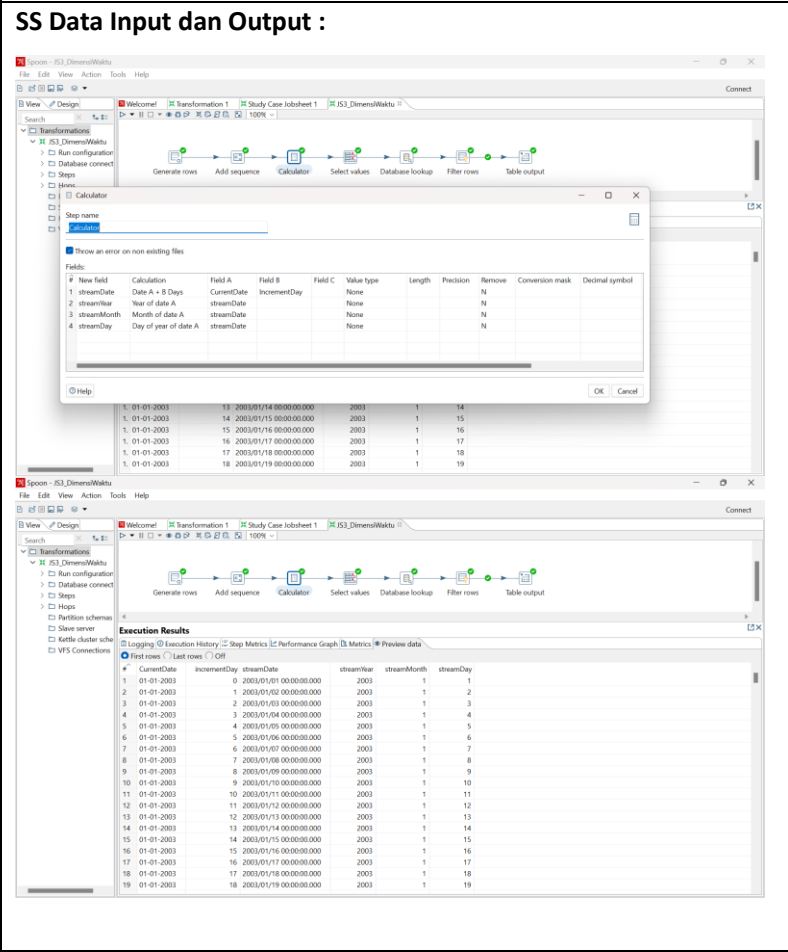




Calculator

Keterangan :

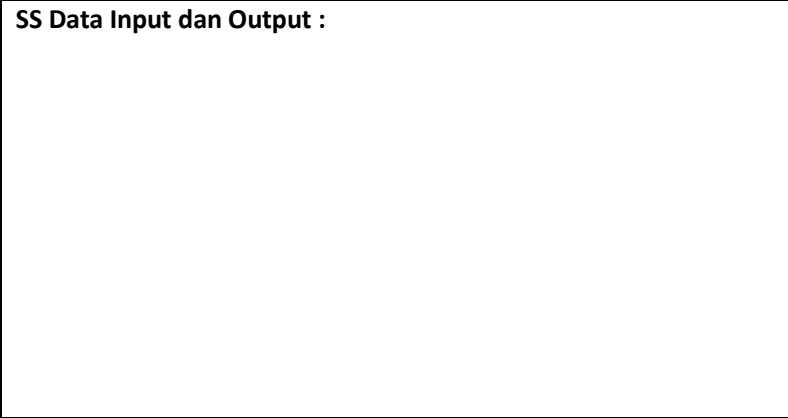
Calculator berguna untuk menghitung data berdasarkan nilai-nilai yang ada. Pada gambar, Calculator digunakan untuk menambahkan kolom baru seperti incrementDay (hasil dari CurrentDate + 1 hari) dan dayOfYear (hari ke berapa dalam satu tahun).



Select values

Keterangan :

Pilih Nilai digunakan untuk memilih, mengganti nama, mengubah tipe data, atau menghapus kolom tertentu dari kumpulan data. Pada gambar, Select Values digunakan untuk



memilih hanya beberapa kolom, seperti CurrentDate dan IncrementDay, dan menyembunyikan kolom lainnya.

The screenshot displays the Kettle (Pentaho) software interface. A 'Select values' dialog box is open, showing a list of fields to be selected or removed. The 'Fields to remove' list is currently empty. The background shows a data flow diagram with steps: Generate rows, Add sequence, Calculator, Select values, Database lookup, Filter rows, and Table output. The 'Execution Results' table at the bottom shows data for streamDate, streamYear, streamMonth, and streamDay.

streamDate	streamYear	streamMonth	streamDay
2003/01/01 00:00:00.000	2003	1	1
2003/01/02 00:00:00.000	2003	1	2
2003/01/03 00:00:00.000	2003	1	3
2003/01/04 00:00:00.000	2003	1	4
2003/01/05 00:00:00.000	2003	1	5
2003/01/06 00:00:00.000	2003	1	6
2003/01/07 00:00:00.000	2003	1	7
2003/01/08 00:00:00.000	2003	1	8
2003/01/09 00:00:00.000	2003	1	9
2003/01/10 00:00:00.000	2003	1	10
2003/01/11 00:00:00.000	2003	1	11
2003/01/12 00:00:00.000	2003	1	12
2003/01/13 00:00:00.000	2003	1	13
2003/01/14 00:00:00.000	2003	1	14
2003/01/15 00:00:00.000	2003	1	15
2003/01/16 00:00:00.000	2003	1	16
2003/01/17 00:00:00.000	2003	1	17
2003/01/18 00:00:00.000	2003	1	18
2003/01/19 00:00:00.000	2003	1	19

Database lookup

Keterangan :

Database Lookup digunakan untuk mencari dan mengambil data dari tabel database berdasarkan nilai kunci tertentu.

SS Data Input dan Output :

Pada gambar, Database Lookup

mengambil data dari skema

dw_legendvehicle dan tabel

dw_dimdate, dengan kolom

streamDate, streamYear,

streamMonth, dan streamDay

sebagai kunci pencarian.

Pencarian mengembalikan kolom

tahun, bulan, dan hari, yang

kemudian digunakan dalam

proses selanjutnya.

The screenshot shows the Kettle Database Lookup configuration window. The step name is 'Database lookup'. The connection is 'conn_dw_definition'. The lookup schema is 'dw_legendvehicle' and the lookup table is 'dimdate'. The cache size is 0. The keys to look up are: 1. date, 2. year, 3. month, 4. day. The values to return are: 1. date, 2. year, 3. month, 4. day. The execution results show a table with columns: streamDate, streamYear, streamMonth, streamDay, year, month, day. The data is as follows:

streamDate	streamYear	streamMonth	streamDay	year	month	day
2003/01/01 00:00:00.000	2003	1	1	<null>	<null>	<null>
2003/01/02 00:00:00.000	2003	1	2	<null>	<null>	<null>
2003/01/03 00:00:00.000	2003	1	3	<null>	<null>	<null>
2003/01/04 00:00:00.000	2003	1	4	<null>	<null>	<null>
2003/01/05 00:00:00.000	2003	1	5	<null>	<null>	<null>
2003/01/06 00:00:00.000	2003	1	6	<null>	<null>	<null>
2003/01/07 00:00:00.000	2003	1	7	<null>	<null>	<null>
2003/01/08 00:00:00.000	2003	1	8	<null>	<null>	<null>
2003/01/09 00:00:00.000	2003	1	9	<null>	<null>	<null>
2003/01/10 00:00:00.000	2003	1	10	<null>	<null>	<null>
2003/01/11 00:00:00.000	2003	1	11	<null>	<null>	<null>
2003/01/12 00:00:00.000	2003	1	12	<null>	<null>	<null>
2003/01/13 00:00:00.000	2003	1	13	<null>	<null>	<null>
2003/01/14 00:00:00.000	2003	1	14	<null>	<null>	<null>
2003/01/15 00:00:00.000	2003	1	15	<null>	<null>	<null>
2003/01/16 00:00:00.000	2003	1	16	<null>	<null>	<null>
2003/01/17 00:00:00.000	2003	1	17	<null>	<null>	<null>
2003/01/18 00:00:00.000	2003	1	18	<null>	<null>	<null>
2003/01/19 00:00:00.000	2003	1	19	<null>	<null>	<null>

Filter rows

Keterangan :

Filter baris digunakan untuk

menyaring data berdasarkan

kondisi tertentu sebelum

disimpan ke tabel output. Filter

Baris diterapkan setelah Database

Lookup yang digunakan untuk

menyaring data dengan kondisi

tertentu.

SS Data Input dan Output :

The screenshot shows the Kettle Filter Rows configuration window. The step name is 'Filter rows'. The condition is: 'date IS NULL'. The execution results show a table with columns: streamDate, streamYear, streamMonth, streamDay, year, month, day. The data is as follows:

streamDate	streamYear	streamMonth	streamDay	year	month	day
2003/01/01 00:00:00.000	2003	1	14	<null>	<null>	<null>
2003/01/15 00:00:00.000	2003	1	15	<null>	<null>	<null>
2003/01/16 00:00:00.000	2003	1	16	<null>	<null>	<null>

Execution Results

#	streamDate	streamYear	streamMonth	streamDay	year	month	day
1	2003/01/01 00:00:00.000	2003	1	1	<null>	<null>	<null>
2	2003/01/02 00:00:00.000	2003	1	2	<null>	<null>	<null>
3	2003/01/03 00:00:00.000	2003	1	3	<null>	<null>	<null>
4	2003/01/04 00:00:00.000	2003	1	4	<null>	<null>	<null>
5	2003/01/05 00:00:00.000	2003	1	5	<null>	<null>	<null>
6	2003/01/06 00:00:00.000	2003	1	6	<null>	<null>	<null>
7	2003/01/07 00:00:00.000	2003	1	7	<null>	<null>	<null>
8	2003/01/08 00:00:00.000	2003	1	8	<null>	<null>	<null>
9	2003/01/09 00:00:00.000	2003	1	9	<null>	<null>	<null>
10	2003/01/10 00:00:00.000	2003	1	10	<null>	<null>	<null>
11	2003/01/11 00:00:00.000	2003	1	11	<null>	<null>	<null>
12	2003/01/12 00:00:00.000	2003	1	12	<null>	<null>	<null>
13	2003/01/13 00:00:00.000	2003	1	13	<null>	<null>	<null>
14	2003/01/14 00:00:00.000	2003	1	14	<null>	<null>	<null>
15	2003/01/15 00:00:00.000	2003	1	15	<null>	<null>	<null>
16	2003/01/16 00:00:00.000	2003	1	16	<null>	<null>	<null>
17	2003/01/17 00:00:00.000	2003	1	17	<null>	<null>	<null>
18	2003/01/18 00:00:00.000	2003	1	18	<null>	<null>	<null>
19	2003/01/19 00:00:00.000	2003	1	19	<null>	<null>	<null>

Table Output

Keterangan :

Tabel Output digunakan untuk menyimpan hasil transformasi data ke dalam tabel tujuan di dalam database. Pada gambar, data konfigurasi akan dimasukkan ke dalam tabel dimdate pada skema dw_legendvehicle dengan menggunakan koneksi conn_dw_destination.

Hasil dari data yang telah dimasukkan ke dalam tabel dimdate pada database menggunakan fitur Table Output.

Data yang tersimpan meliputi kolom id_dimDate, tanggal, tahun, bulan, dan hari. Setiap baris menampilkan tanggal dari Januari 2003, dengan nilai tahun, bulan, dan hari yang sesuai. Hal ini menunjukkan bahwa proses penyimpanan data telah berhasil dilakukan dengan menggunakan integrasi data Pentaho dan

SS Data Input dan Output :

Execution Results

#	streamDate	streamYear	streamMonth	streamDay	year	month	day
1	2003/01/01 00:00:00.000	2003	1	1	<null>	<null>	<null>
2	2003/01/02 00:00:00.000	2003	1	2	<null>	<null>	<null>
3	2003/01/03 00:00:00.000	2003	1	3	<null>	<null>	<null>
4	2003/01/04 00:00:00.000	2003	1	4	<null>	<null>	<null>
5	2003/01/05 00:00:00.000	2003	1	5	<null>	<null>	<null>
6	2003/01/06 00:00:00.000	2003	1	6	<null>	<null>	<null>
7	2003/01/07 00:00:00.000	2003	1	7	<null>	<null>	<null>
8	2003/01/08 00:00:00.000	2003	1	8	<null>	<null>	<null>
9	2003/01/09 00:00:00.000	2003	1	9	<null>	<null>	<null>
10	2003/01/10 00:00:00.000	2003	1	10	<null>	<null>	<null>
11	2003/01/11 00:00:00.000	2003	1	11	<null>	<null>	<null>
12	2003/01/12 00:00:00.000	2003	1	12	<null>	<null>	<null>
13	2003/01/13 00:00:00.000	2003	1	13	<null>	<null>	<null>
14	2003/01/14 00:00:00.000	2003	1	14	<null>	<null>	<null>
15	2003/01/15 00:00:00.000	2003	1	15	<null>	<null>	<null>
16	2003/01/16 00:00:00.000	2003	1	16	<null>	<null>	<null>
17	2003/01/17 00:00:00.000	2003	1	17	<null>	<null>	<null>
18	2003/01/18 00:00:00.000	2003	1	18	<null>	<null>	<null>
19	2003/01/19 00:00:00.000	2003	1	19	<null>	<null>	<null>

dikelola dengan database
(phpMyAdmin).

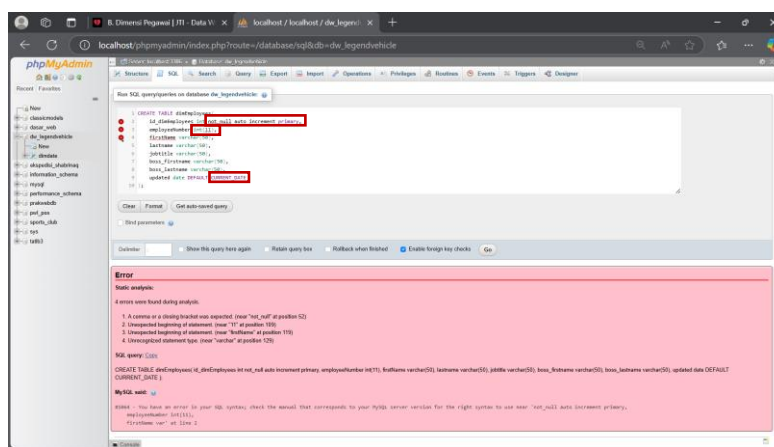
		id_dimDate	date	year	month	day
☐	 Edit  Copy  Delete	1	2003-01-01	2003	1	1
☐	 Edit  Copy  Delete	2	2003-01-02	2003	1	2
☐	 Edit  Copy  Delete	3	2003-01-03	2003	1	3
☐	 Edit  Copy  Delete	4	2003-01-04	2003	1	4
☐	 Edit  Copy  Delete	5	2003-01-05	2003	1	5
☐	 Edit  Copy  Delete	6	2003-01-06	2003	1	6
☐	 Edit  Copy  Delete	7	2003-01-07	2003	1	7
☐	 Edit  Copy  Delete	8	2003-01-08	2003	1	8
☐	 Edit  Copy  Delete	9	2003-01-09	2003	1	9
☐	 Edit  Copy  Delete	10	2003-01-10	2003	1	10
☐	 Edit  Copy  Delete	11	2003-01-11	2003	1	11
☐	 Edit  Copy  Delete	12	2003-01-12	2003	1	12
☐	 Edit  Copy  Delete	13	2003-01-13	2003	1	13
☐	 Edit  Copy  Delete	14	2003-01-14	2003	1	14
☐	 Edit  Copy  Delete	15	2003-01-15	2003	1	15
☐	 Edit  Copy  Delete	16	2003-01-16	2003	1	16
☐	 Edit  Copy  Delete	17	2003-01-17	2003	1	17
☐	 Edit  Copy  Delete	18	2003-01-18	2003	1	18
☐	 Edit  Copy  Delete	19	2003-01-19	2003	1	19
☐	 Edit  Copy  Delete	20	2003-01-20	2003	1	20
☐	 Edit  Copy  Delete	21	2003-01-21	2003	1	21
☐	 Edit  Copy  Delete	22	2003-01-22	2003	1	22
☐	 Edit  Copy  Delete	23	2003-01-23	2003	1	23
☐	 Edit  Copy  Delete	24	2003-01-24	2003	1	24
☐	 Edit  Copy  Delete	25	2003-01-25	2003	1	25
 ☐ Check all With selected:  Edit  Copy  Delete  Export						

B. Dimensi Pegawai

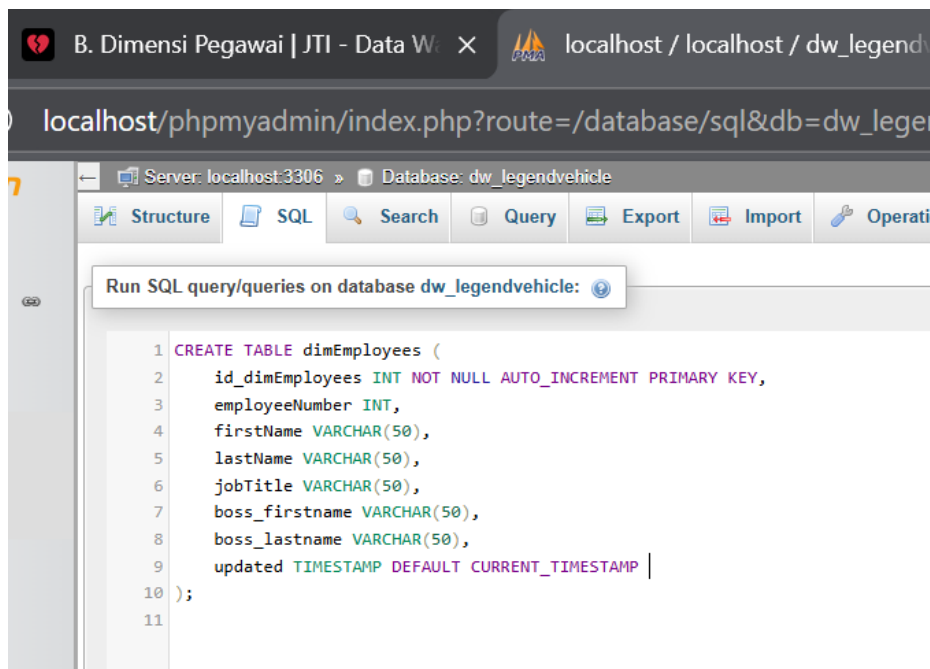
1. Buatlah tabel **dimPegawai** pada **dw_legendVehicle**.

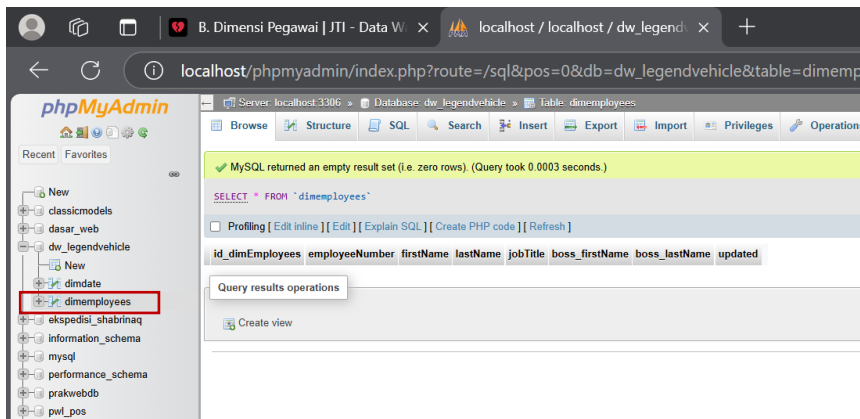
```
CREATE TABLE dimEmployees(  
  id_dimEmployees int not_null auto increment primary,  
  employeeNumber int(11),  
  firstName varchar(50),  
  lastName varchar(50),  
  jobtitle varchar(50),  
  boss_firstname varchar(50),  
  boss_lastname varchar(50),  
  updated date DEFAULT CURRENT_DATE  
);
```

My Answer :



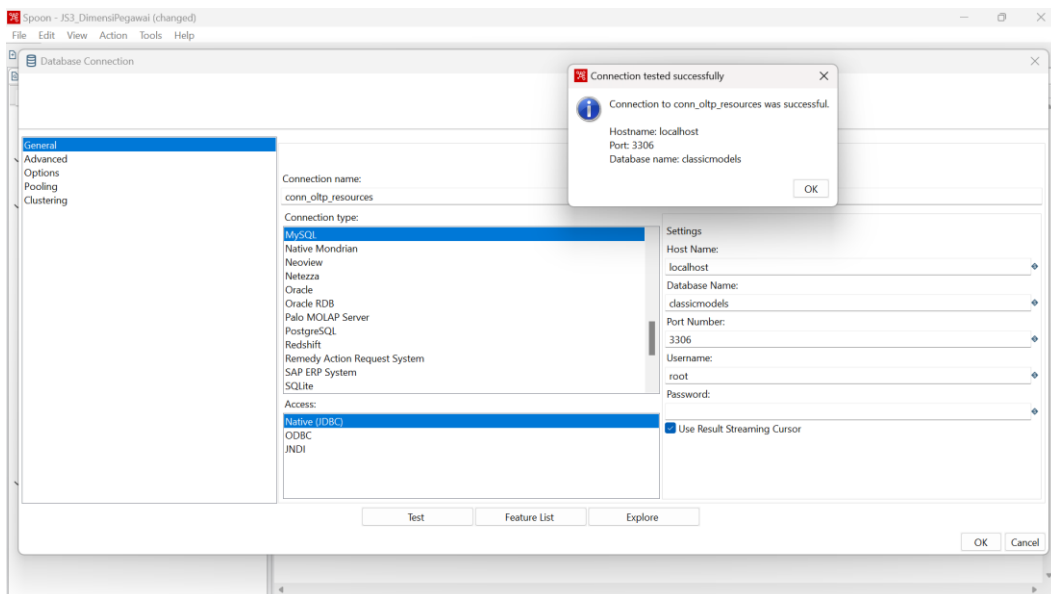
perbaiki kode



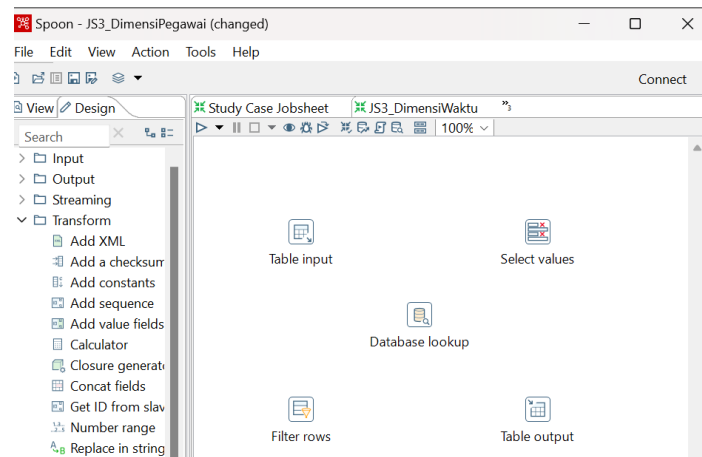


2. Pada PDI Spoon buatlah koneksi baru dengan nama **conn_oltp_resources** yang menghubungkan dengan database oltp. sesuaikan **hostname**, **database name**, **port number**, **username** dan **password** dengan keadaan pada device masing-masing.

My Answer :



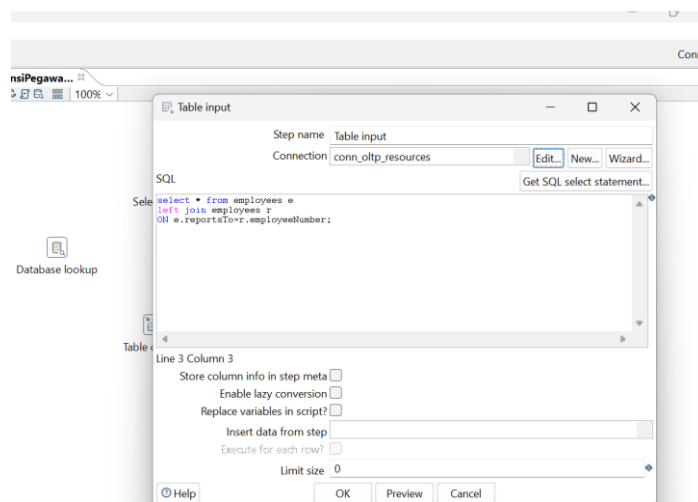
3. Drag and drop beberapa objek sebagai berikut:
 - **Table input:** digunakan mengambil data dari database OLTP.
 - **Select values:** memilih field yang digunakan untuk proses Transform dan Load.
 - **Database lookup:** digunakan untuk melihat data pada tabel dimEmployees untuk memastikan data tidak kembar.
 - **Filter rows:** digunakan untuk memilih data stream yang masih belum ada apada tabel dimEmployees.
 - **Table output:** Memasukkan data ke dalam table dimEmployees



4. Konfigurasi pada table input dengan menghubungkan **Connection** pada koneksi **conn_oltp_resources**. Untuk mengambil data sumber menggunakan query dibawah ini.

```
select * from employees e
left join employees r
on e.reportsTo=r.employeeNumber;
```

My Answer :



Examine preview data

Rows of step: Table input (23 rows)

#	employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle	employeeNumber_1	lastName_1	firstName_1	extension_1	email_1
1	1002	Murphy	Diane	x5800	dmurphy@classicmodelcars.com	1	<null>	President	<null>	<null>	<null>	<null>	<null>
2	1056	Patterson	Mary	x4611	mpatterson@classicmodelcars.com	1	1002	VP Sales	1002	Murphy	Diane	x5800	dmurph
3	1076	Firelli	Jeff	x9273	jfirell@classicmodelcars.com	1	1002	VP Marketing	1002	Murphy	Diane	x5800	dmurph
4	1088	Patterson	William	x4871	wpatterson@classicmodelcars.com	6	1056	Sales Manager (APAC)	1056	Patterson	Mary	x4611	mpatter
5	1102	Bondur	Gerard	x5408	gbondur@classicmodelcars.com	4	1056	Sale Manager (EMEA)	1056	Patterson	Mary	x4611	mpatter
6	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)	1056	Patterson	Mary	x4611	mpatter
7	1165	Jennings	Leslie	x3291	ljennings@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@
8	1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@
9	1188	Firelli	Julie	x2173	jfirell@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@
1	1216	Patterson	Steve	x4334	spatterson@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@
1	1286	Tseng	Foon Yue	x2248	ftseng@classicmodelcars.com	3	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@
1	1323	Vanauf	George	x4102	gvanauf@classicmodelcars.com	3	1143	Sales Rep	1143	Bow	Anthony	x5428	abow@
1	1337	Bondur	Loui	x6493	lbondur@classicmodelcars.com	4	1102	Sales Rep	1102	Bondur	Gerard	x5408	gbondu
1	1370	Hernandez	Gerard	x2028	ghernande@classicmodelcars.com	4	1102	Sales Rep	1102	Bondur	Gerard	x5408	gbondu
1	1401	Castillo	Pamela	x2759	pcastillo@classicmodelcars.com	4	1102	Sales Rep	1102	Bondur	Gerard	x5408	gbondu
1	1501	Rott	Larry	x2311	lrott@classicmodelcars.com	7	1102	Sales Rep	1102	Bondur	Gerard	x5408	gbondu
1	1504	Jones	Barry	x102	bjones@classicmodelcars.com	7	1102	Sales Rep	1102	Bondur	Gerard	x5408	gbondu
1	1611	Foster	Andy	x101	afoster@classicmodelcars.com	6	1088	Sales Rep	1088	Patterson	William	x4871	wpatt
1	1612	Marsh	Peter	x102	pmarsh@classicmodelcars.com	6	1088	Sales Rep	1088	Patterson	William	x4871	wpatt
2	1619	King	Tom	x103	tking@classicmodelcars.com	6	1088	Sales Rep	1088	Patterson	William	x4871	wpatt
2	1621	Nishi	Mami	x101	mnishi@classicmodelcars.com	5	1056	Sales Rep	1056	Patterson	Mary	x4611	mpatter
2	1625	Kato	Yoshimi	x102	ykato@classicmodelcars.com	5	1621	Sales Rep	1621	Nishi	Mami	x101	mnishi
2	1702	Gerard	Martin	x2312	mgerard@classicmodelcars.com	4	1102	Sales Rep	1102	Bondur	Gerard	x5408	gbondu

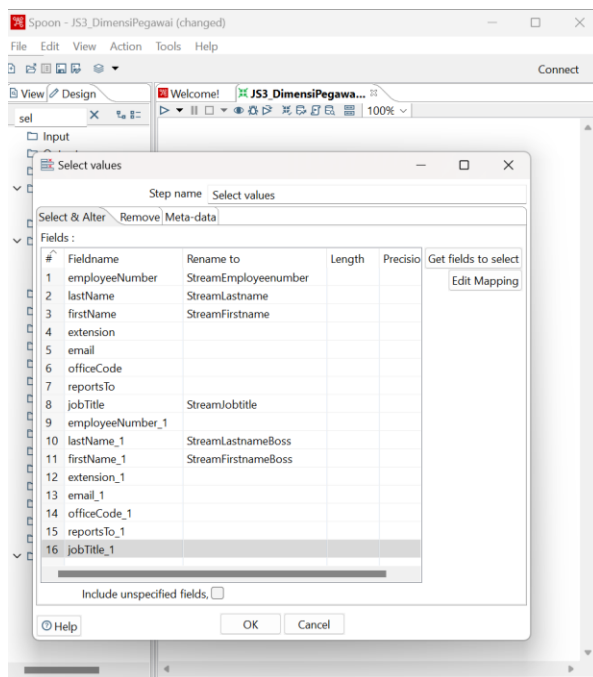
5. Hubungkan output **table input** pada **select values**.

My Answer :



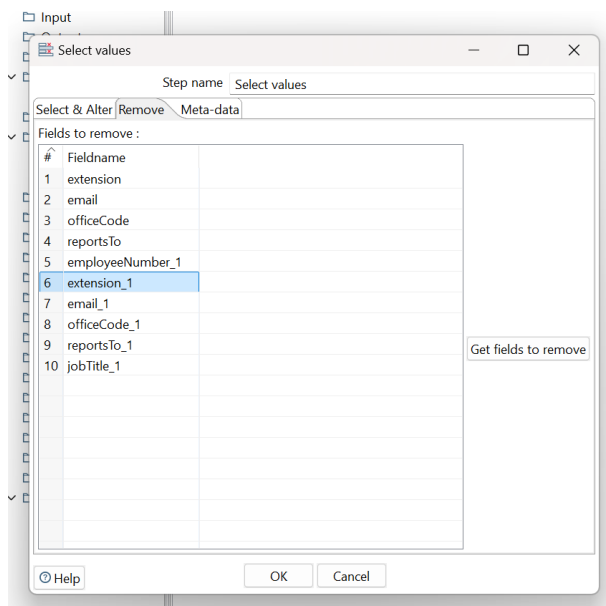
6. Konfigurasi pada **Select values** yaitu mengambil data dari field **employeenumber, lastname, firstname, jobtitle, lastname_1 dan firstname_1** sebagai **data stream** yang digunakan pada proses ETL pada tab **select & alter**.

My Answer :



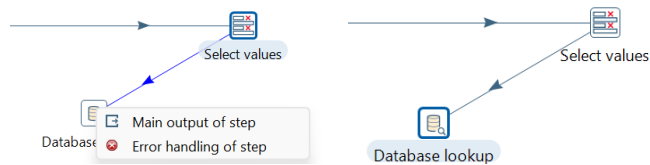
7. Hilangkan field lain yang tidak digunakan pada tab **remove**.

My Answer :



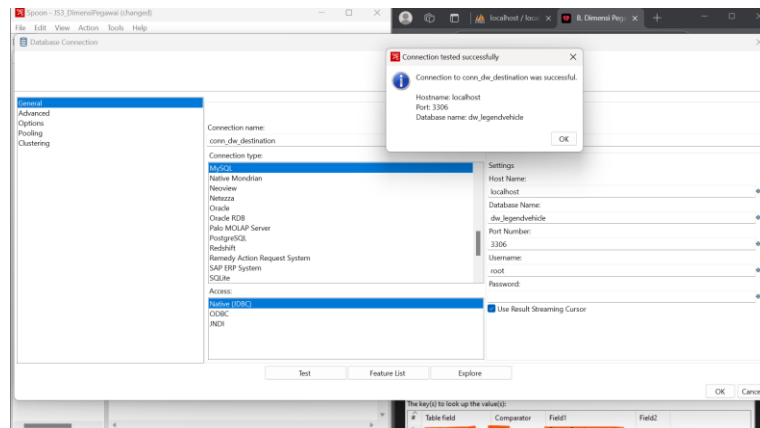
8. Hubungkan output **select values** pada **database lookup**

My Answer :



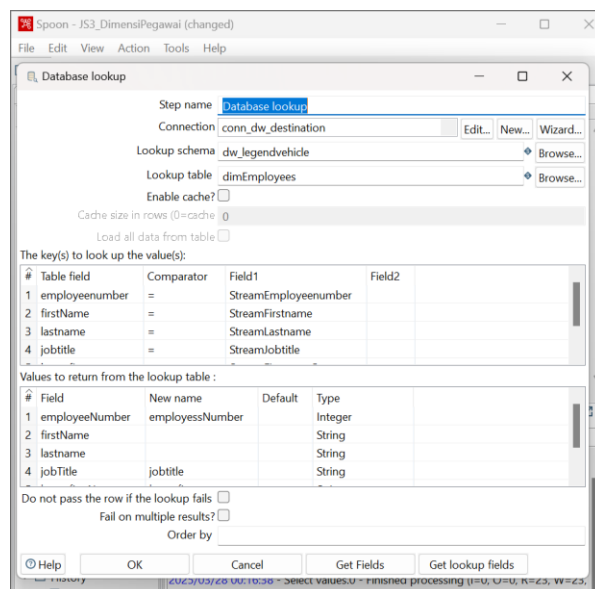
9. Konfigurasi pada database lookup adalah dengan menghubungkan koneksi pada **conn_dw_destination** dengan table lookup **dimEmployees** yang telah dibuat pada tahap pertama.

My Answer :



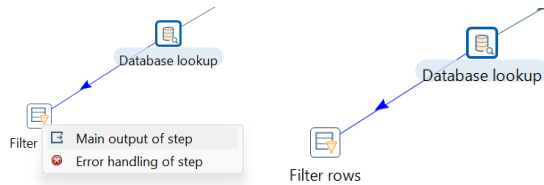
10. Field yang di lookup adalah field pada tabel **dimEmployees** dengan **field stream input** dari OLTP. sedangkan field yang di **retrieve** adalah field dari **dimEmployees** itu sendiri. Jika tidak ada data yang sama maka akan muncul null.

My Answer :



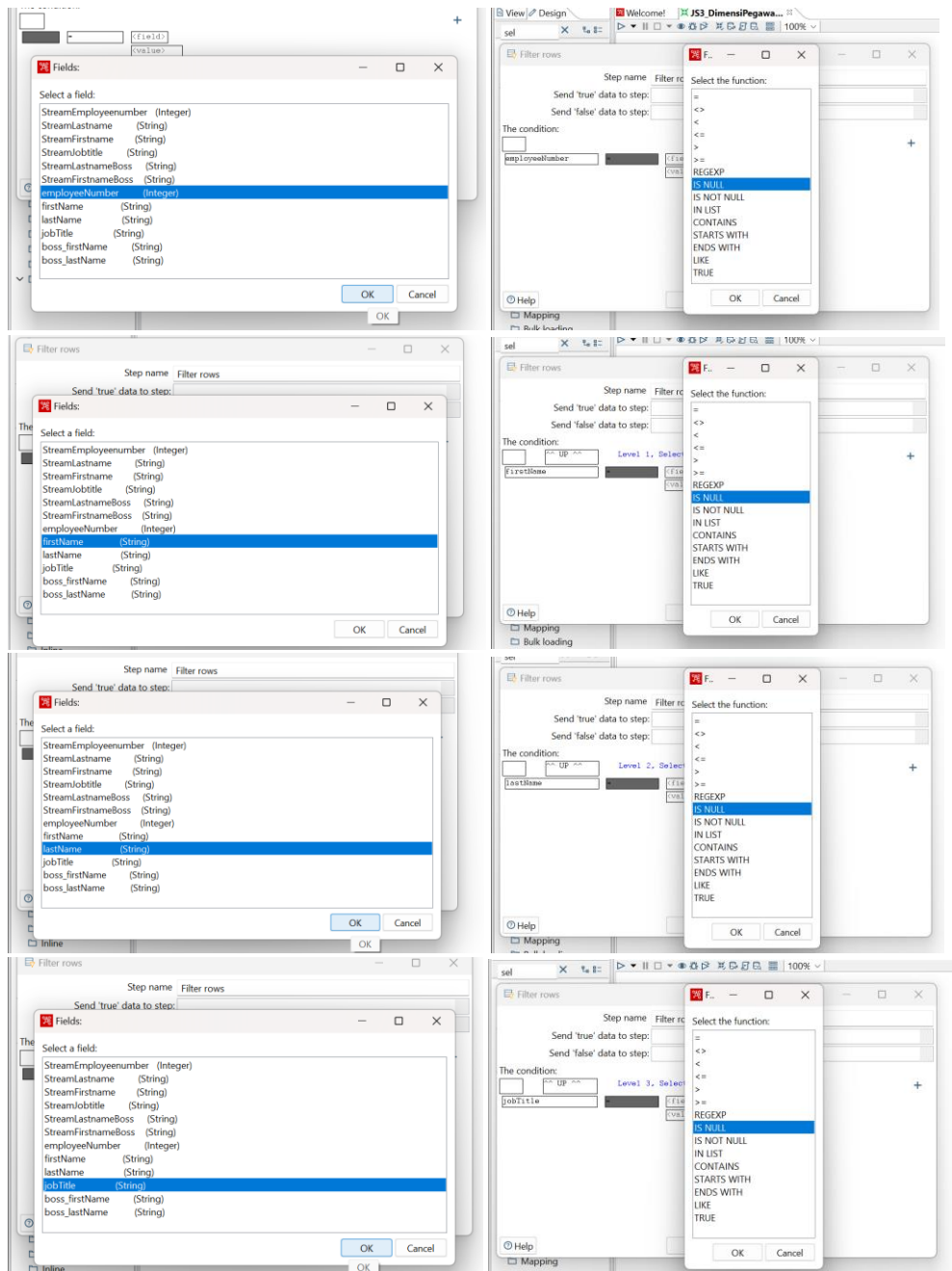
11. Hubungkan **output database lookup** dengan **filter rows**.

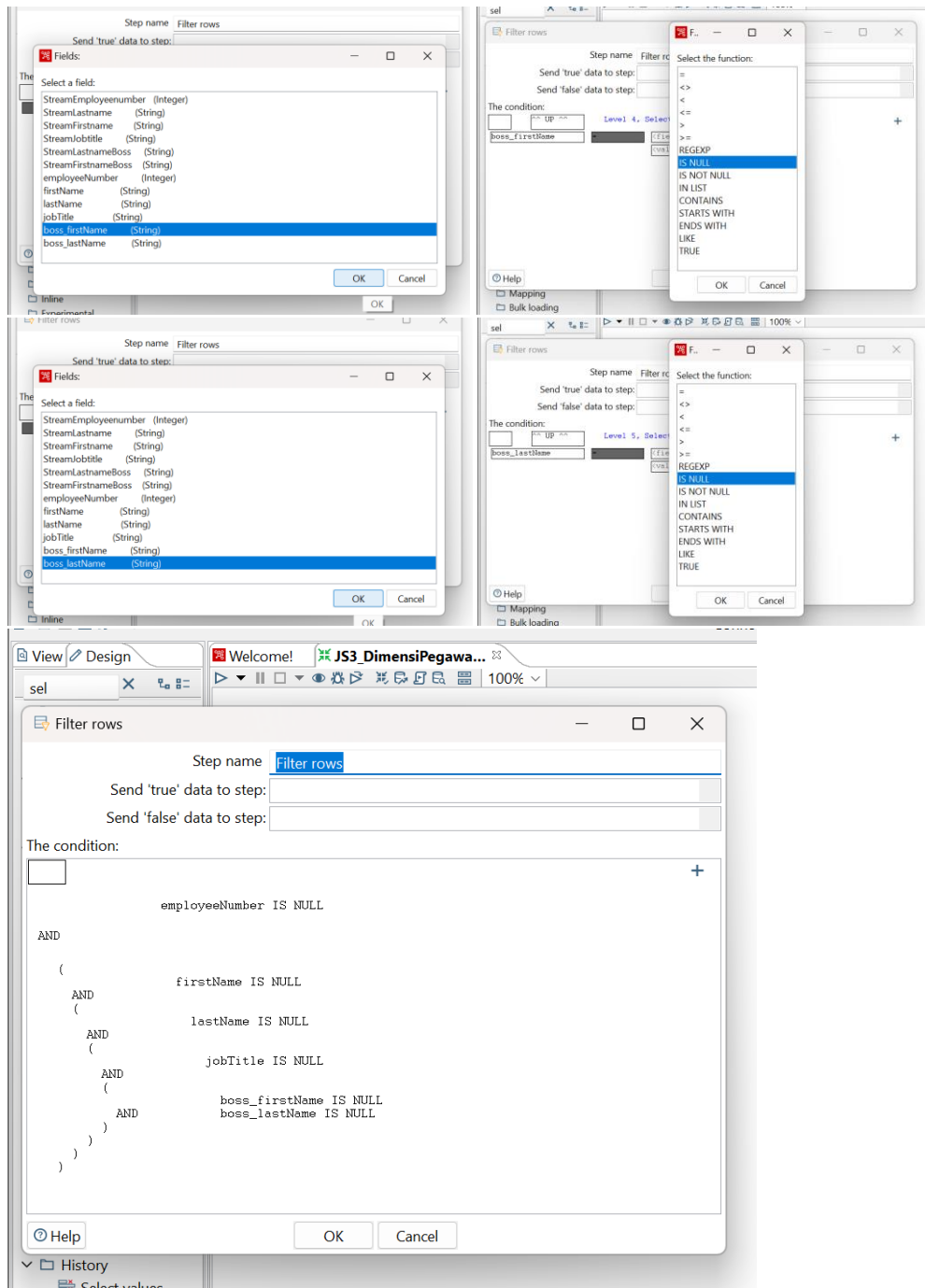
My Answer :



12. Pada **filter rows** berikan kondisi field yang **null** pada **field dimemployees** untuk dimasukkan pada proses selanjutnya. Hal itu menandakan bahwa **data stream** belum memiliki kesamaan pada data di **dimemployees**.

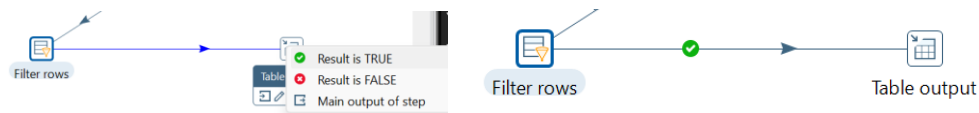
My Answer :





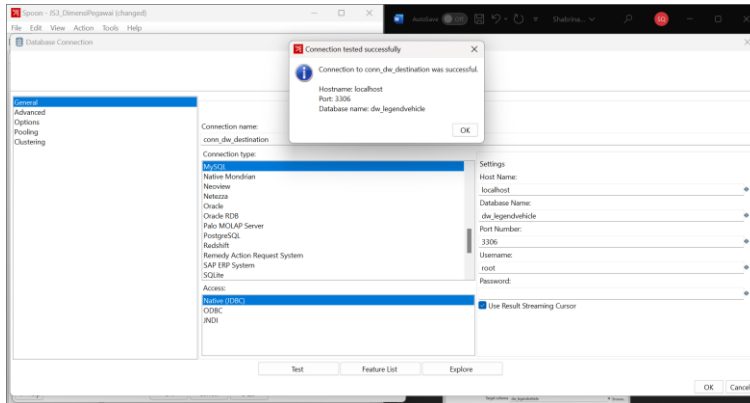
13. Hubungkan output dari **filter rows** dengan **table output**.

My Answer :



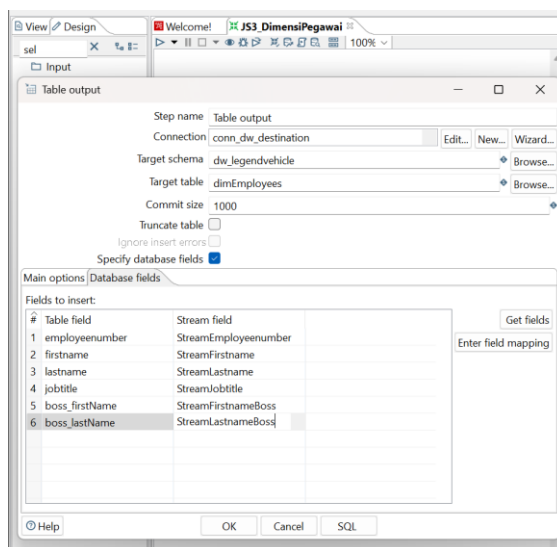
14. Pada **table output**, gunakan **connection conn_dw_destination** untuk memasukkan data pada tabel **dimemployees**.

My Answer :



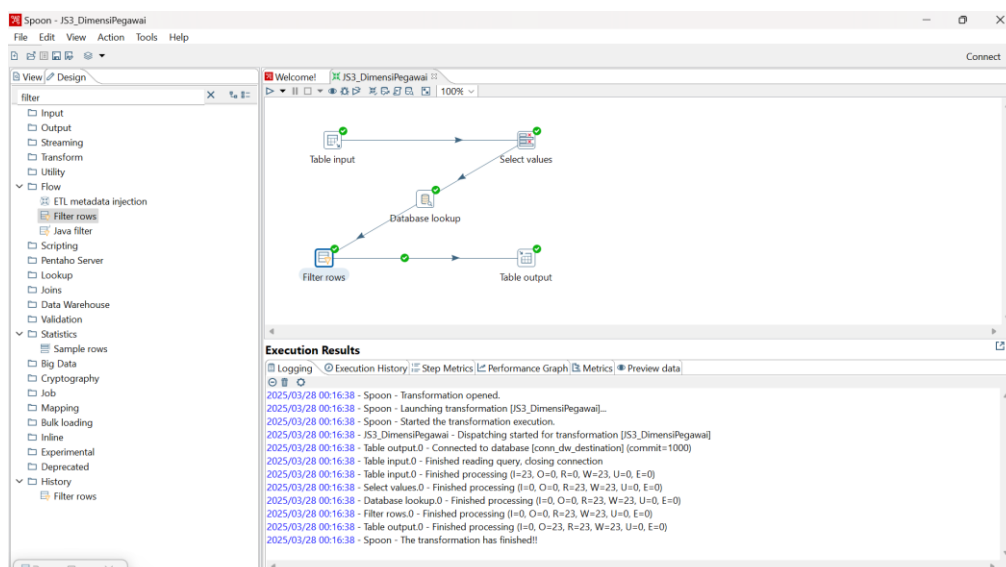
15. Aktifkan **specify database fields**, dan mapping **data stream input** dari oltp terhadap field yang ada pada **dimemployees**.

My Answer :



16. jika proses keseluruhan berhasil maka tabel **dimemployees** akan terisi data pegawai dari database OLTP.

My Answer :



Spoon - JS3_DimensiPegawai (changed)

File Edit View Action Tools Help

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Table input

Database lookup

Filter rows

Select values

Table output

Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

First rows Last rows Off

#	employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle	employeeNumber_1	lastName_1	firstName_1
1	1002	Murphy	Diane	x5800	dmurphy@classicmodelcars.com	1	<null>	President	<null>	<null>	<null>
2	1056	Patterson	Mary	x4611	mpatterson@classicmodelcars.com	1	1002	VP Sales	1002	Murphy	Diane
3	1076	Finelli	Jeff	x9273	jfinelli@classicmodelcars.com	1	1002	VP Marketing	1002	Murphy	Diane
4	1088	Patterson	William	x4871	wpatterson@classicmodelcars.com	6	1056	Sales Manager (APAC)	1056	Patterson	Mary
5	1102	Bondur	Gerard	x5408	gbondur@classicmodelcars.com	4	1056	Sales Manager (EMEA)	1056	Patterson	Mary
6	1143	Bow	Anthony	x5428	abow@classicmodelcars.com	1	1056	Sales Manager (NA)	1056	Patterson	Mary
7	1165	Jennings	Leslie	x3291	ljennings@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony
8	1166	Thompson	Leslie	x4065	lthompson@classicmodelcars.com	1	1143	Sales Rep	1143	Bow	Anthony
9	1188	Finelli	Julie	x2173	jfinelli@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony
1.	1216	Patterson	Steve	x4334	spatterson@classicmodelcars.com	2	1143	Sales Rep	1143	Bow	Anthony
1.	1286	Tseng	Foon Yue	x2248	ftseng@classicmodelcars.com	3	1143	Sales Rep	1143	Bow	Anthony
1.	1323	Vanauf	George	x4102	gvanauf@classicmodelcars.com	3	1143	Sales Rep	1143	Bow	Anthony
1.	1337	Bondur	Loui	x6493	lbondur@classicmodelcars.com	4	1102	Sales Rep	1102	Bondur	Gerard

Spoon - JS3_DimensiPegawai (changed)

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History
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Filter rows

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Table output

Execution Results

Logging Execution History Step Metrics Performance Graph Metrics Preview data

First rows Last rows Off

#	StreamEmployeeNumber	StreamLastName	StreamFirstName	StreamJobTitle	StreamLastNameBoss	StreamFirstNameBoss
1	1002	Murphy	Diane	President	<null>	<null>
2	1056	Patterson	Mary	VP Sales	Murphy	Diane
3	1076	Finelli	Jeff	VP Marketing	Murphy	Diane
4	1088	Patterson	William	Sales Manager (APAC)	Patterson	Mary
5	1102	Bondur	Gerard	Sale Manager (EMEA)	Patterson	Mary
6	1143	Bow	Anthony	Sales Manager (NA)	Patterson	Mary
7	1165	Jennings	Leslie	Sales Rep	Bow	Anthony
8	1166	Thompson	Leslie	Sales Rep	Bow	Anthony
9	1188	Finelli	Julie	Sales Rep	Bow	Anthony
1.	1216	Patterson	Steve	Sales Rep	Bow	Anthony
1.	1286	Tseng	Foon Yue	Sales Rep	Bow	Anthony
1.	1323	Vanauf	George	Sales Rep	Bow	Anthony
1.	1337	Bondur	Loui	Sales Rep	Bondur	Gerard
1.	1370	Hernandez	Gerard	Sales Rep	Bondur	Gerard

Spoon - JS3_DimensiPegawai (changed)

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#	StreamEmployeeNumber	StreamLastName	StreamFirstName	StreamJobTitle	StreamLastNameBoss	StreamFirstNameBoss	employessNumber	firstName	lastName	jobtitle	bo
1	1002	Murphy	Diane	President	<null>	<null>	<null>	<null>	<null>	<null>	<null>
2	1056	Patterson	Mary	VP Sales	Murphy	Diane	<null>	<null>	<null>	<null>	<null>
3	1076	Finelli	Jeff	VP Marketing	Murphy	Diane	<null>	<null>	<null>	<null>	<null>
4	1088	Patterson	William	Sales Manager (APAC)	Patterson	Mary	<null>	<null>	<null>	<null>	<null>
5	1102	Bondur	Gerard	Sale Manager (EMEA)	Patterson	Mary	<null>	<null>	<null>	<null>	<null>
6	1143	Bow	Anthony	Sales Manager (NA)	Patterson	Mary	<null>	<null>	<null>	<null>	<null>
7	1165	Jennings	Leslie	Sales Rep	Bow	Anthony	<null>	<null>	<null>	<null>	<null>
8	1166	Thompson	Leslie	Sales Rep	Bow	Anthony	<null>	<null>	<null>	<null>	<null>
9	1188	Finelli	Julie	Sales Rep	Bow	Anthony	<null>	<null>	<null>	<null>	<null>
1.	1216	Patterson	Steve	Sales Rep	Bow	Anthony	<null>	<null>	<null>	<null>	<null>
1.	1286	Tseng	Foon Yue	Sales Rep	Bow	Anthony	<null>	<null>	<null>	<null>	<null>
1.	1323	Vanauf	George	Sales Rep	Bow	Anthony	<null>	<null>	<null>	<null>	<null>
1.	1337	Bondur	Loui	Sales Rep	Bondur	Gerard	<null>	<null>	<null>	<null>	<null>

employeeNumber	firstName	lastName	jobTitle	boxes_firstName	boxes_lastName	updated
1	Diane	Murphy	President	NIJEL	NIJEL	2025-03-28 00:16:38
2	Mary	Patterson	VP Sales	Diane	Murphy	2025-03-28 00:16:38
3	Jeff	Firelli	VP Marketing	Diane	Murphy	2025-03-28 00:16:38
4	William	Patterson	Sales Manager (APAC)	Mary	Patterson	2025-03-28 00:16:38
5	Gerard	Bondur	Sales Manager (EMEA)	Mary	Patterson	2025-03-28 00:16:38
6	Anthony	Bow	Sales Manager (NA)	Mary	Patterson	2025-03-28 00:16:38
7	Leslie	Jennings	Sales Rep	Anthony	Bow	2025-03-28 00:16:38
8	Leslie	Thompson	Sales Rep	Anthony	Bow	2025-03-28 00:16:38
9	Julie	Firelli	Sales Rep	Anthony	Bow	2025-03-28 00:16:38
10	Steve	Patterson	Sales Rep	Anthony	Bow	2025-03-28 00:16:38
11	Foon Yue	Tsang	Sales Rep	Anthony	Bow	2025-03-28 00:16:38
12	George	Vanauf	Sales Rep	Anthony	Bow	2025-03-28 00:16:38
13	Loui	Bondur	Sales Rep	Gerard	Bondur	2025-03-28 00:16:38
14	Gerard	Hernandez	Sales Rep	Gerard	Bondur	2025-03-28 00:16:38
15	Pamela	Castillo	Sales Rep	Gerard	Bondur	2025-03-28 00:16:38
16	Larry	Butt	Sales Rep	Gerard	Bondur	2025-03-28 00:16:38
17	Barry	Jones	Sales Rep	Gerard	Bondur	2025-03-28 00:16:38
18	Andy	Flater	Sales Rep	William	Patterson	2025-03-28 00:16:38
19	Peter	Marsh	Sales Rep	William	Patterson	2025-03-28 00:16:38
20	Tom	King	Sales Rep	William	Patterson	2025-03-28 00:16:38

TUGAS 2

1. Buka preview tab pada execution result area di setiap proses object. amati input dan output data yang ada. bandingkan di setiap prosesnya. jelaskan perbedaan di setiap prosesnya.

Table Input

Keterangan :

Step Table Input digunakan untuk mengambil data karyawan dari database (saya mengambil datanya dari database classicmodels yang merupakan database jobsheet sebelumnya) dengan query SQL. Outputnya berupa tabel yang menampilkan data karyawan beserta informasi atasannya untuk diproses lebih lanjut.

SS Data Input dan Output :

Step name: **Table input**
 Connection: **conn_oltp_resources**

SQL:

```
select * from employees e
left join employees r
ON e.reportsTo=r.employeeNumber;
```

Line 1 Column 0

Store column info in step meta ☐
 Enable lazy conversion ☐
 Replace variables in script? ☐
 Insert data from step ☐
 Execute for each row? ☐
 Limit size: 0

Preview data:

employeeNumber	lastName	extension	email	officeCode	reportsTo	jobTitle	employeeNumber_1	lastName_1	extension_1	email_1
1	Murphy	x5800	dmurphy@classicmodels.com	1	<null>	President	<null>	<null>	<null>	<null>
2	Patterson	x4611	mpatterson@classicmodels.com	1	1002	VP Sales	1002	Murphy	Diane	x5800
3	Firelli	x2773	jfirelli@classicmodels.com	1	1002	VP Marketing	1002	Murphy	Diane	x5800
4	Patterson	x4871	vpatterson@classicmodels.com	6	1056	Sales Manager (APAC)	1056	Patterson	Mary	x4611
5	Bondur	x5408	gbondur@classicmodels.com	4	1056	Sales Manager (EMEA)	1056	Patterson	Mary	x4611
6	Bow	x5428	abow@classicmodels.com	1	1056	Sales Manager (NA)	1056	Patterson	Mary	x4611
7	Jennings	x3291	ljennings@classicmodels.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428
8	Thompson	x4065	lthompson@classicmodels.com	1	1143	Sales Rep	1143	Bow	Anthony	x5428
9	Firelli	x2773	jfirelli@classicmodels.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428
10	Patterson	x4334	spatterson@classicmodels.com	2	1143	Sales Rep	1143	Bow	Anthony	x5428
11	Tsang	x2248	ftsang@classicmodels.com	3	1143	Sales Rep	1143	Bow	Anthony	x5428
12	Vanauf	x4102	gvanauf@classicmodels.com	3	1143	Sales Rep	1143	Bow	Anthony	x5428
13	Bondur	x5493	lbondur@classicmodels.com	4	1102	Sales Rep	1102	Bondur	Gerard	x5408
14	Hernandez	x2038	ghernandez@classicmodels.com	7	1102	Sales Rep	1102	Bondur	Gerard	x5408
15	Castillo	x2759	pcastillo@classicmodels.com	4	1102	Sales Rep	1102	Bondur	Gerard	x5408
16	Butt	x2311	lbutt@classicmodels.com	7	1102	Sales Rep	1102	Bondur	Gerard	x5408
17	Jones	x1102	ljones@classicmodels.com	4	1102	Sales Rep	1102	Bondur	Gerard	x5408
18	Flater	x101	aflater@classicmodels.com	6	1088	Sales Rep	1088	Patterson	William	x4871
19	Marsh	x102	pmarsh@classicmodels.com	6	1088	Sales Rep	1088	Patterson	William	x4871
20	King	x103	tking@classicmodels.com	6	1088	Sales Rep	1088	Patterson	William	x4871
21	Nishi	x101	mnishi@classicmodels.com	5	1056	Sales Rep	1056	Patterson	Mary	x4611
22	Kato	x102	ykato@classicmodels.com	5	1621	Nails	1621	Nails	x101	mnishi@classicmodels.com
23	Gerard	x2312	mgard@classicmodels.com	4	1102	Sales Rep	1102	Bondur	Gerard	x5408

Select Values

Keterangan :

Step Select Values digunakan untuk memilih, mengganti nama, atau mengubah tipe data kolom yang diambil dari sumber data (database). Outputnya adalah data dengan kolom yang telah dimodifikasi sesuai kebutuhan proses selanjutnya.

SS Data Input dan Output :

The 'Select values' dialog box shows the mapping of source fields to target fields. The 'Fields' tab is active, displaying a list of fields with their original names, renamed names, lengths, and precisions. The 'Include unspecified fields' checkbox is unchecked.

The 'Execution Results' table shows the output of the 'Select values' step. The table has columns for 'StreamEmployeeNumber', 'StreamLastname', 'StreamFirstname', 'StreamJobtitle', 'StreamLastnameBoss', and 'StreamFirstnameBoss'. The data is organized into rows, with some rows having a '1.' prefix, indicating a specific instance of the data.

#	StreamEmployeeNumber	StreamLastname	StreamFirstname	StreamJobtitle	StreamLastnameBoss	StreamFirstnameBoss
1	1002	Murphy	Diane	President	<null>	<null>
2	1056	Patterson	Mary	VP Sales	Murphy	Diane
3	1076	Firrelli	Jeff	VP Marketing	Murphy	Diane
4	1088	Patterson	William	Sales Manager (APAC)	Patterson	Mary
5	1102	Bondur	Gerard	Sale Manager (EMEA)	Patterson	Mary
6	1143	Bow	Anthony	Sales Manager (NA)	Patterson	Mary
7	1165	Jennings	Leslie	Sales Rep	Bow	Anthony
8	1166	Thompson	Leslie	Sales Rep	Bow	Anthony
9	1188	Firrelli	Julie	Sales Rep	Bow	Anthony
1.	1216	Patterson	Steve	Sales Rep	Bow	Anthony
1.	1286	Tseng	Foon Yue	Sales Rep	Bow	Anthony
1.	1323	Vanauf	George	Sales Rep	Bow	Anthony
1.	1337	Bondur	Loui	Sales Rep	Bondur	Gerard
1.	1370	Hernandez	Gerard	Sales Rep	Bondur	Gerard
1.	1401	Castillo	Pamela	Sales Rep	Bondur	Gerard
1.	1501	Bott	Larry	Sales Rep	Bondur	Gerard
1.	1504	Jones	Barry	Sales Rep	Bondur	Gerard
1.	1611	Fixter	Andy	Sales Rep	Patterson	William
1.	1612	Marsh	Peter	Sales Rep	Patterson	William
2.	1619	King	Tom	Sales Rep	Patterson	William
2.	1621	Nishi	Mami	Sales Rep	Patterson	Mary
2.	1625	Kato	Yoshimi	Sales Rep	Nishi	Mami

Database Lookup

Keterangan :

Step Database Lookup digunakan untuk mengambil data tambahan dari tabel lain dalam database berdasarkan key value tertentu. Outputnya merupakan data informasi tambahan yang telah dicocokkan dari tabel referensi.

SS Data Input dan Output :

The 'Database lookup' dialog box shows the configuration for a database lookup step. The 'Step name' is 'Database lookup'. The 'Connection' is 'conn_dw_destination'. The 'Lookup schema' is 'dw_legendvehicle'. The 'Lookup table' is 'dimEmployees'. The 'Enable cache?' checkbox is unchecked. The 'Cache size in rows (0=cache)' is set to 0. The 'Load all data from table' checkbox is unchecked.

The 'The key(s) to look up the value(s):' section shows a table with columns for 'Table field', 'Comparator', 'Field1', and 'Field2'. The data is organized into rows, with some rows having a '1.' prefix, indicating a specific instance of the data.

#	Table field	Comparator	Field1	Field2
1	employeenumber	=	StreamEmployeeNumber	
2	firstName	=	StreamFirstname	
3	lastName	=	StreamLastname	
4	jobtitle	=	StreamJobtitle	

The 'Values to return from the lookup table:' section shows a table with columns for 'Field', 'New name', 'Default', and 'Type'. The data is organized into rows, with some rows having a '1.' prefix, indicating a specific instance of the data.

#	Field	New name	Default	Type
1	employeeNumber	employessNumber		Integer
2	firstName			String
3	lastName			String
4	jobTitle	jobtitle		String

The 'Do not pass the row if the lookup fails' checkbox is unchecked. The 'Fail on multiple results?' checkbox is unchecked. The 'Order by' field is empty.

#	Name	Type	Collat
☐ 1	id_dimEmployees	int	
☐ 2	employeeNumber	int	
☐ 3	firstName	varchar(50)	utf8mt
☐ 4	lastName	varchar(50)	utf8mt
☐ 5	jobTitle	varchar(50)	utf8mt
☐ 6	boss_firstName	varchar(50)	utf8mt
☐ 7	boss_lastName	varchar(50)	utf8mt

Filter Rows

Keterangan :

Step Filter Rows digunakan untuk menyaring data berdasarkan kondisi tertentu. Outputnya merupakan data yang memenuhi kondisi modifikasi yang dikirim ke satu jalur dan data yang tidak memenuhi dikirim ke jalur lain. filter rows diterapkan setelah tabel database lookup.

SS Data Input dan Output :

Filter rows

Step name:

Send 'true' data to step:

Send 'false' data to step:

The condition:

☐

Filter rows

Step name:

Send 'true' data to step:

Send 'false' data to step:

The condition:

```

employeeNumber IS NULL
AND
(
  AND
  (
    AND
    (
      AND
      (
        AND
        (
          boss_firstName IS NULL
          AND
          boss_lastName IS NULL
        )
      )
    )
  )
  AND
  (
    firstName IS NULL
    AND
    lastName IS NULL
    AND
    jobTitle IS NULL
  )
)

```

OK Cancel

Table Output

Keterangan :

Step Table Output digunakan untuk menyimpan data hasil transformasi ke dalam tabel database. Outputnya merupakan data yang telah diproses dan dimasukkan ke dalam tabel tujuan sesuai dengan konfigurasi yang ditentukan. data

SS Data Input dan Output :

Table output

Step name:

Connection:

Target schema:

Target table:

Commit size:

Truncate table: ☐

Specify database fields: ☒

Main options / Database fields

#	Table field	Stream field
1	employeeNumber	StreamEmployeeNumber
2	firstName	StreamFirstName
3	lastName	StreamLastName
4	jobTitle	StreamJobTitle
5	boss_firstName	StreamFirstNameBoss
6	boss_lastName	StreamLastNameBoss

OK Cancel SQL

hasil output tabel output akan muncul di database (phpMyAdmin).

The screenshot shows the phpMyAdmin interface with the 'dimemployees' table selected. The table structure is as follows:

id_dimemployees	employeeNumber	firstName	lastName	jobTitle	boxes_firstName	boxes_lastName	updated
1	1002	Diane	Murphy	President	NULL	NULL	2025-03-20 00:16:30
2	1056	Mary	Patterson	VP Sales	Diane	Murphy	2025-03-20 00:16:30
3	1076	Jeff	Finelli	VP Marketing	Diane	Murphy	2025-03-20 00:16:30
4	1088	William	Patterson	Sales Manager (APAC)	Mary	Patterson	2025-03-20 00:16:30
5	1102	Gerard	Bondur	Sales Manager (EMEA)	Mary	Patterson	2025-03-20 00:16:30
6	1143	Anthony	Bow	Sales Manager (NA)	Mary	Patterson	2025-03-20 00:16:30
7	1165	Leslie	Jennings	Sales Rep	Anthony	Bow	2025-03-20 00:16:30
8	1166	Leslie	Thompson	Sales Rep	Anthony	Bow	2025-03-20 00:16:30
9	1188	Julie	Finelli	Sales Rep	Anthony	Bow	2025-03-20 00:16:30
10	1216	Steve	Patterson	Sales Rep	Anthony	Bow	2025-03-20 00:16:30
11	1286	Foon Yuen	Tsang	Sales Rep	Anthony	Bow	2025-03-20 00:16:30
12	1323	George	Vanaud	Sales Rep	Anthony	Bow	2025-03-20 00:16:30
13	1337	Lois	Bondur	Sales Rep	Gerard	Bondur	2025-03-20 00:16:30
14	1370	Gerard	Hernandez	Sales Rep	Gerard	Bondur	2025-03-20 00:16:30
15	1401	Pamela	Castillo	Sales Rep	Gerard	Bondur	2025-03-20 00:16:30

- Jika proses itu di ulangi (di run kembali) apakah data akan redundan?

My Answer :

Ya, bisa terjadi redundant. Mengapa? karena jika kita insert kembali proses tersebut tanpa adanya pencegahan atau validasi maka Ketika kita mencoba run ulang, data yang mau kita insert kembali akan masuk tanpa menghapus data yang lama, sehingga akan terjadi redundant atau duplikasi.

- Tambahkan nama anda pada table employee di OLTP. jalankan kembali transformasi ini. Amati hasilnya, apa yang terjadi?

My Answer :

The screenshot shows the phpMyAdmin interface with the 'dimemployees' table. A new row has been added at the bottom of the table, highlighted with a red box:

id_dimemployees	employeeNumber	firstName	lastName	jobTitle	boxes_firstName	boxes_lastName	updated
1	1002	Diane	Murphy	President	NULL	NULL	2025-03-20 00:16:30
2	1056	Mary	Patterson	VP Sales	Diane	Murphy	2025-03-20 00:16:30
3	1076	Jeff	Finelli	VP Marketing	Diane	Murphy	2025-03-20 00:16:30
4	1088	William	Patterson	Sales Manager (APAC)	Mary	Patterson	2025-03-20 00:16:30
5	1102	Gerard	Bondur	Sales Manager (EMEA)	Mary	Patterson	2025-03-20 00:16:30
6	1143	Anthony	Bow	Sales Manager (NA)	Mary	Patterson	2025-03-20 00:16:30
7	1165	Leslie	Jennings	Sales Rep	Anthony	Bow	2025-03-20 00:16:30
8	1166	Leslie	Thompson	Sales Rep	Anthony	Bow	2025-03-20 00:16:30
9	1188	Julie	Finelli	Sales Rep	Anthony	Bow	2025-03-20 00:16:30
10	1216	Steve	Patterson	Sales Rep	Anthony	Bow	2025-03-20 00:16:30
11	1286	Foon Yuen	Tsang	Sales Rep	Anthony	Bow	2025-03-20 00:16:30
12	1323	George	Vanaud	Sales Rep	Anthony	Bow	2025-03-20 00:16:30
13	1337	Lois	Bondur	Sales Rep	Gerard	Bondur	2025-03-20 00:16:30
14	1370	Gerard	Hernandez	Sales Rep	Gerard	Bondur	2025-03-20 00:16:30
15	1401	Pamela	Castillo	Sales Rep	Gerard	Bondur	2025-03-20 00:16:30
16	1501	Larry	Bull	Sales Rep	Gerard	Bondur	2025-03-20 00:16:30
17	1504	Barry	Jones	Sales Rep	Gerard	Bondur	2025-03-20 00:16:30
18	1611	Andy	Fisher	Sales Rep	William	Patterson	2025-03-20 00:16:30
19	1612	Peter	March	Sales Rep	William	Patterson	2025-03-20 00:16:30
20	1619	Tom	King	Sales Rep	William	Patterson	2025-03-20 00:16:30
21	1621	Mami	Nishi	Sales Rep	Mary	Patterson	2025-03-20 00:16:30
22	1625	Yoshimi	Kato	Sales Rep	Mami	Nishi	2025-03-20 00:16:30
23	1702	Martin	Gerard	Sales Rep	Gerard	Bondur	2025-03-20 00:16:30
24	1718	Shabina	Guthrie	Business Analyst	Kelly	Sellars	2025-03-20 01:10:09

Ketika saya insert data saya ke table tersebut, maka hasilnya data yang saya masukkan berhasil dan ada pada tabel tersebut.